

BFS Traversal

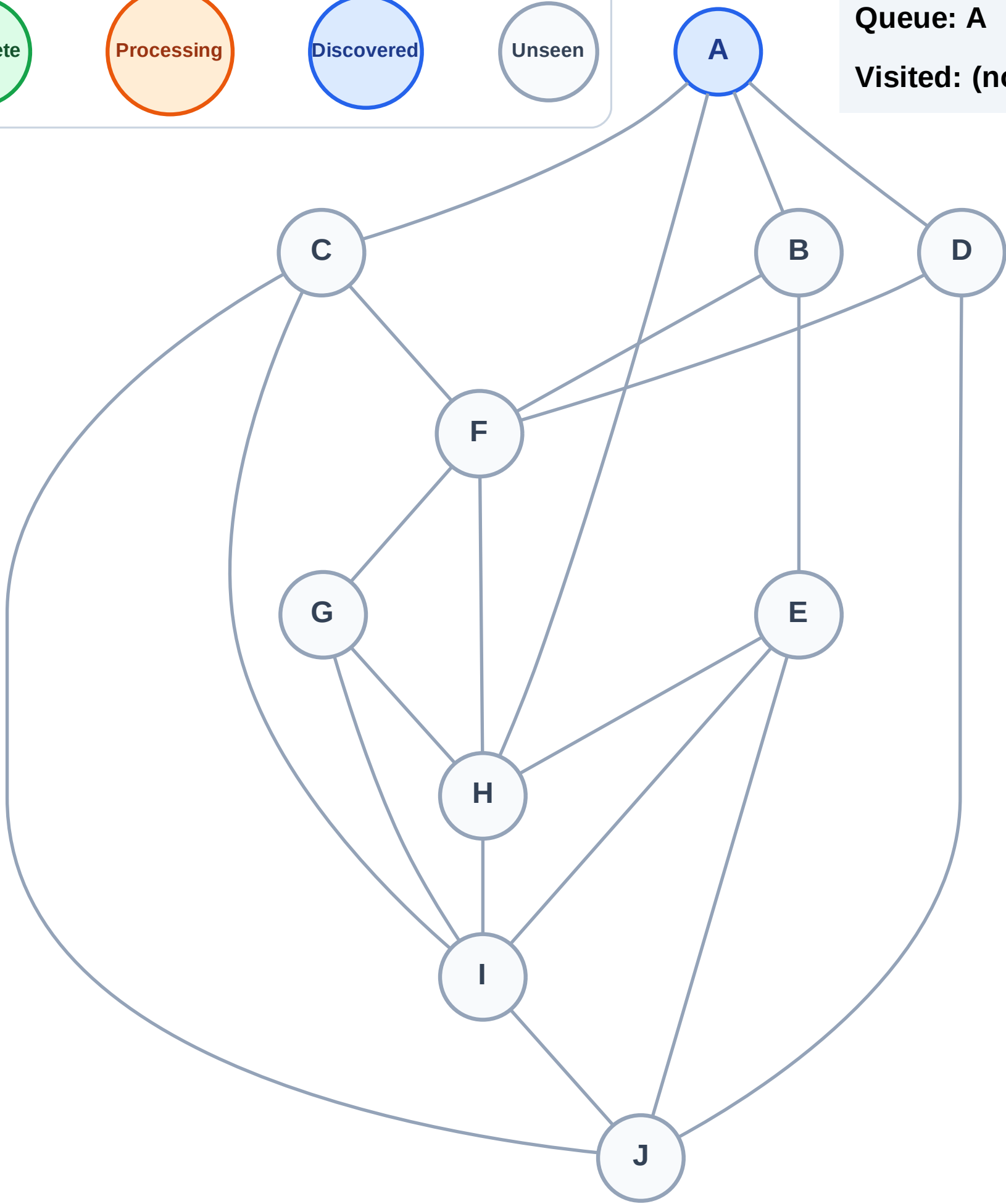
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

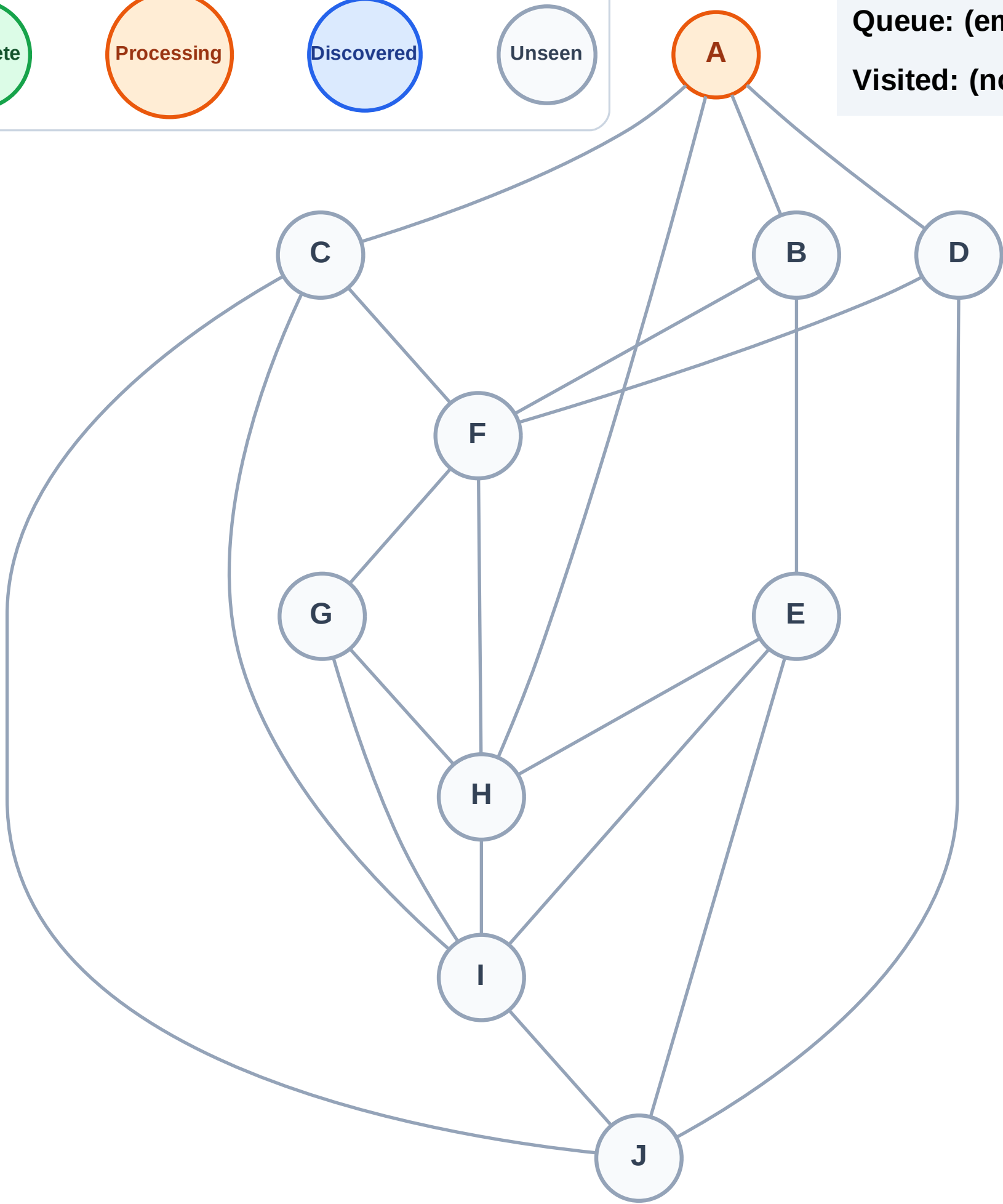
Step 2: Process node A

Legend



Queue: (empty)

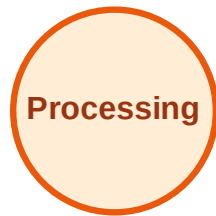
Visited: (none)



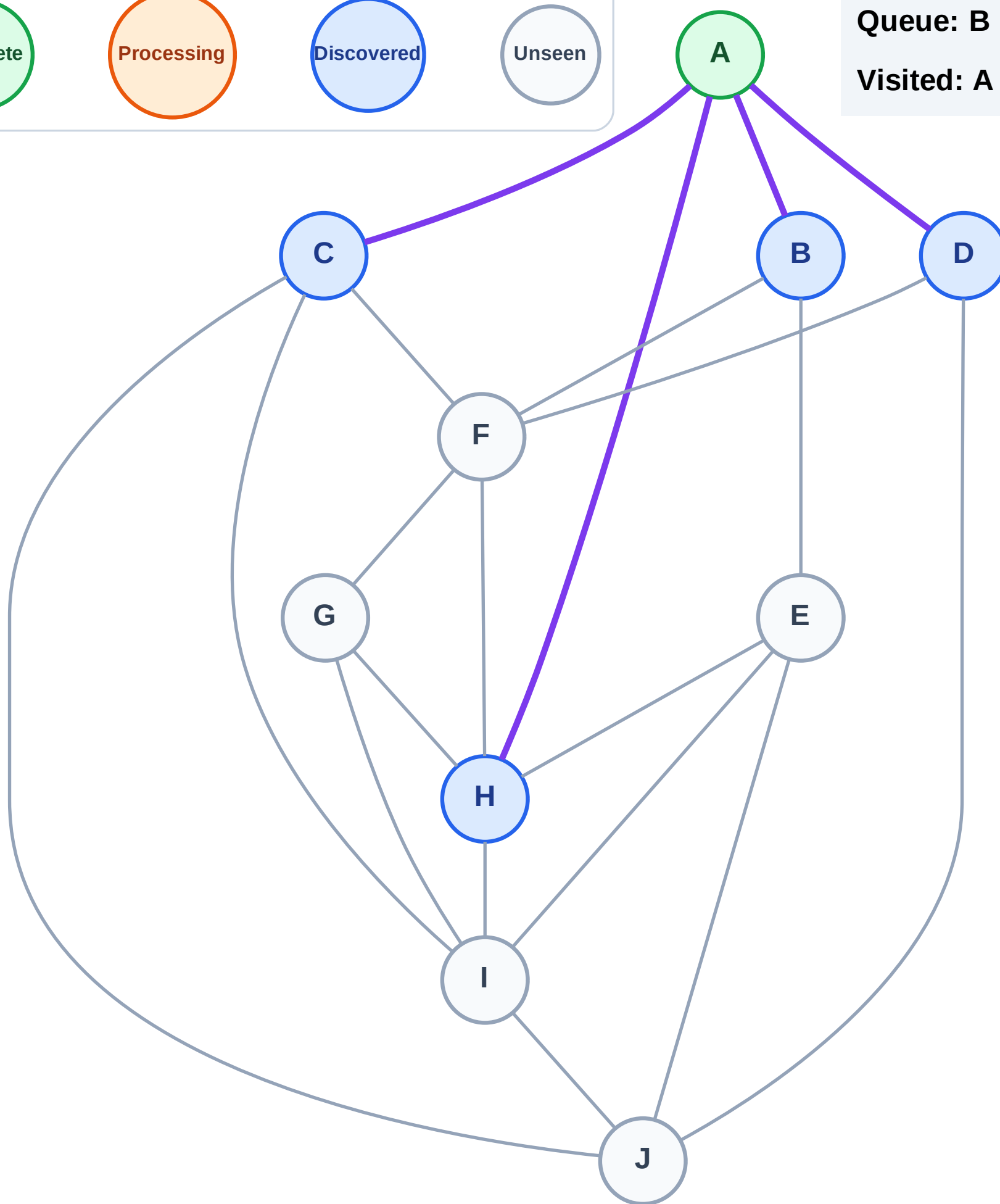
BFS Traversal

Step 3: Discover B, C, D, H from A

Legend



Queue: B → C → D → H
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

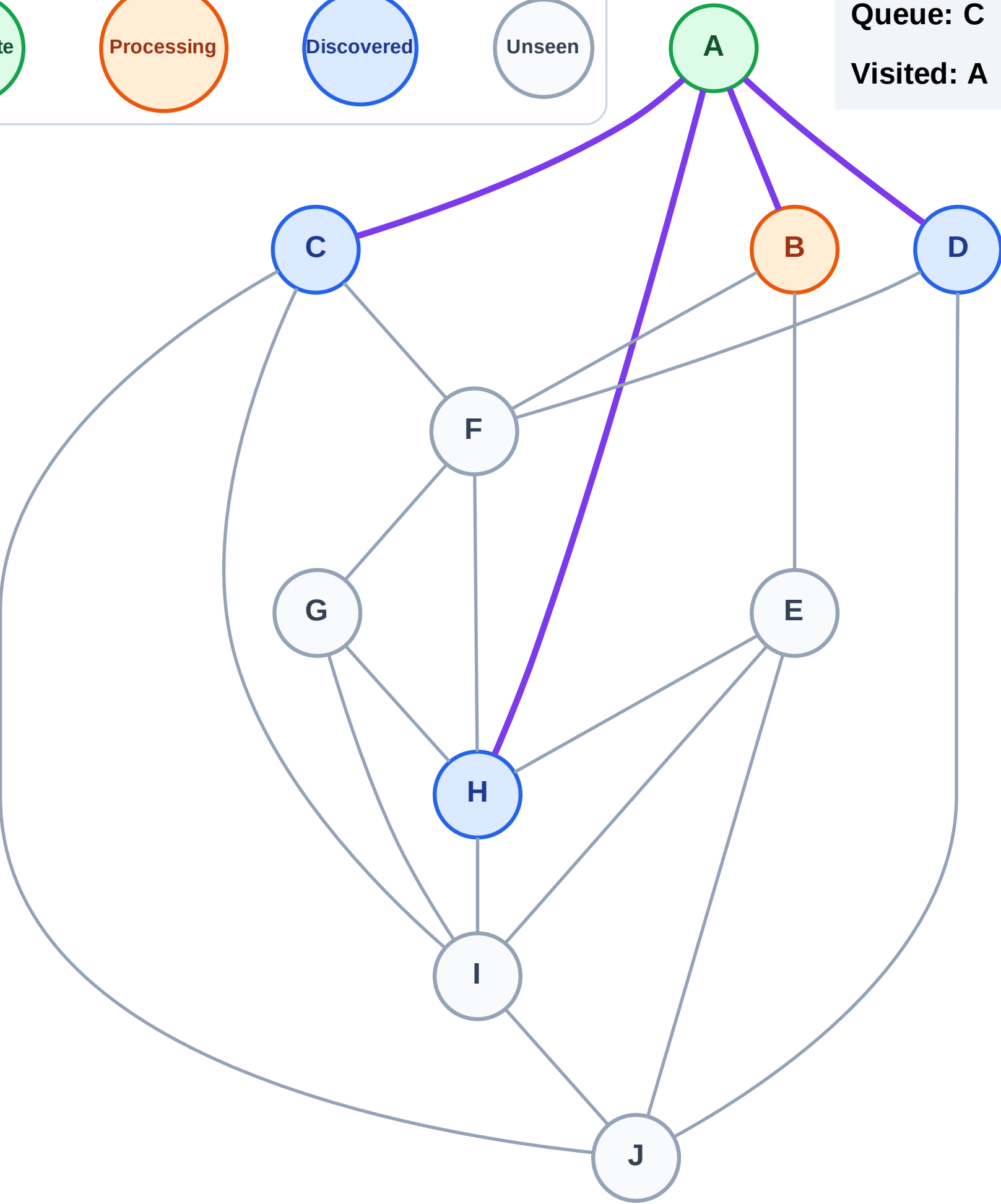
Processing

Discovered

Unseen

Queue: C → D → H

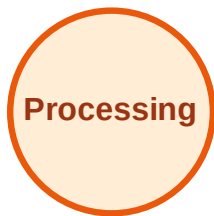
Visited: A



BFS Traversal

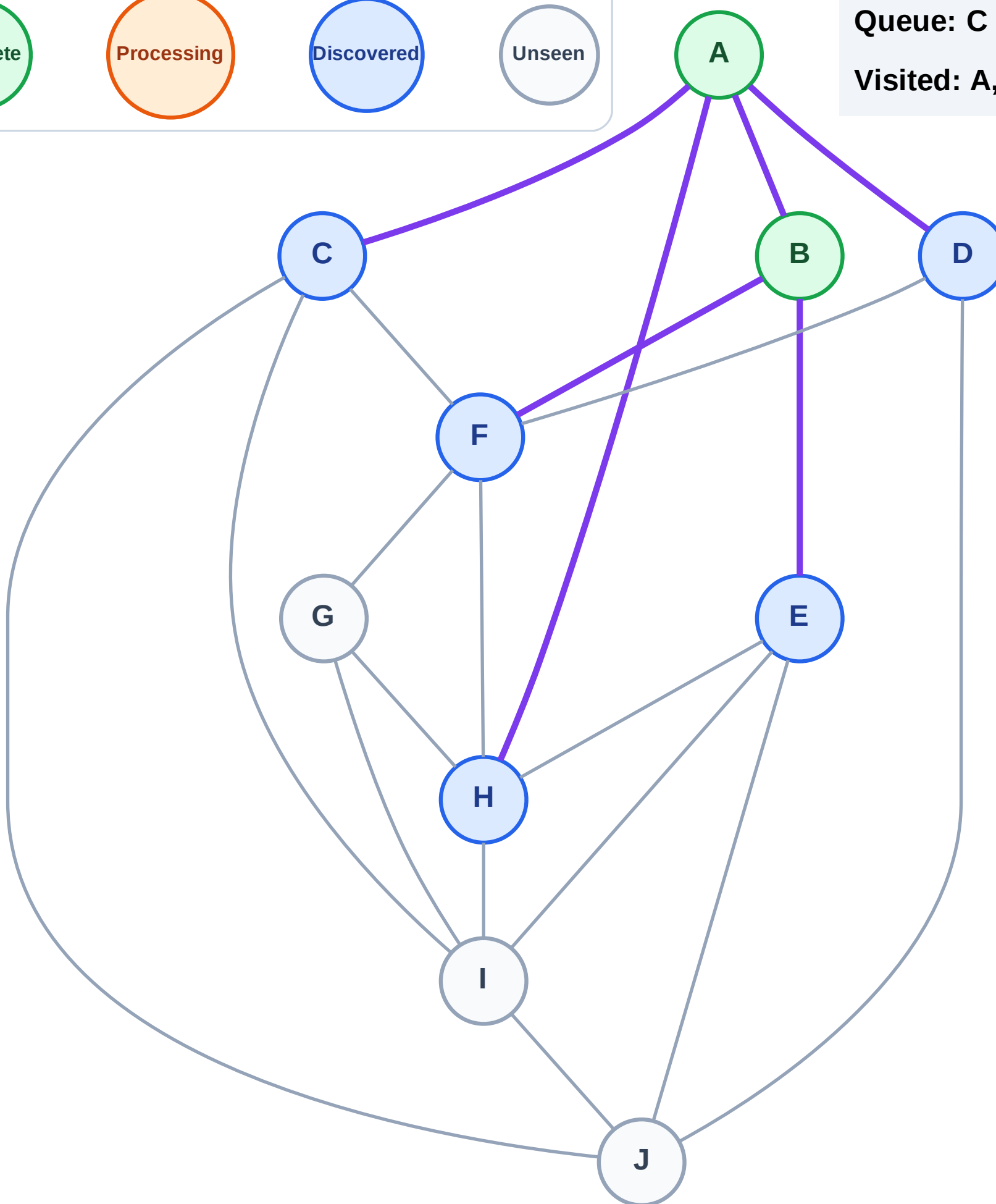
Step 5: Discover E, F from B

Legend



Queue: C → D → H → E → F

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

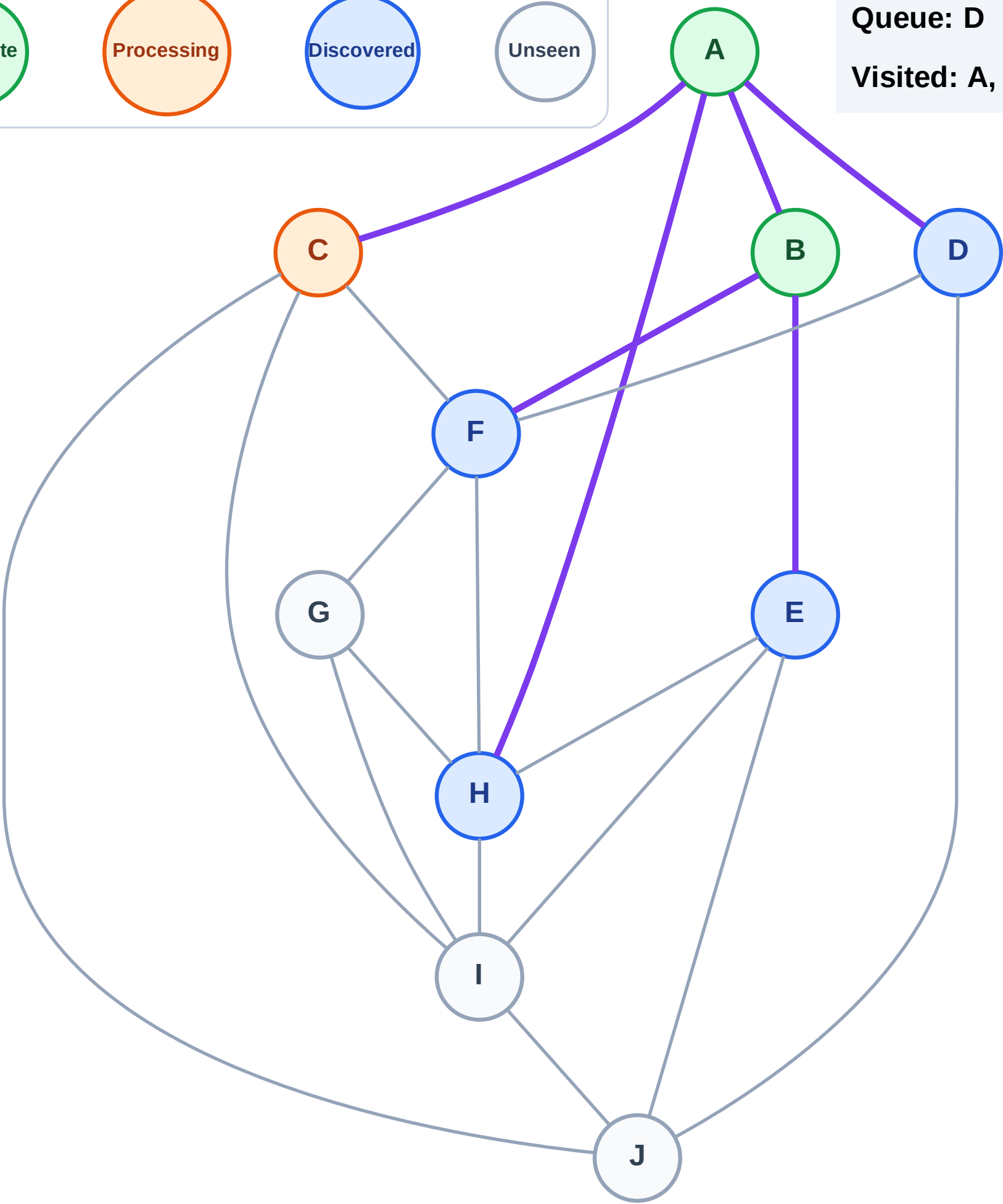
Processing

Discovered

Unseen

Queue: D → H → E → F

Visited: A, B



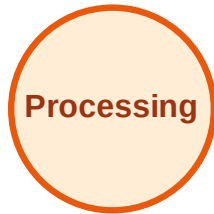
BFS Traversal

Step 7: Discover I, J from C

Legend



Complete



Processing

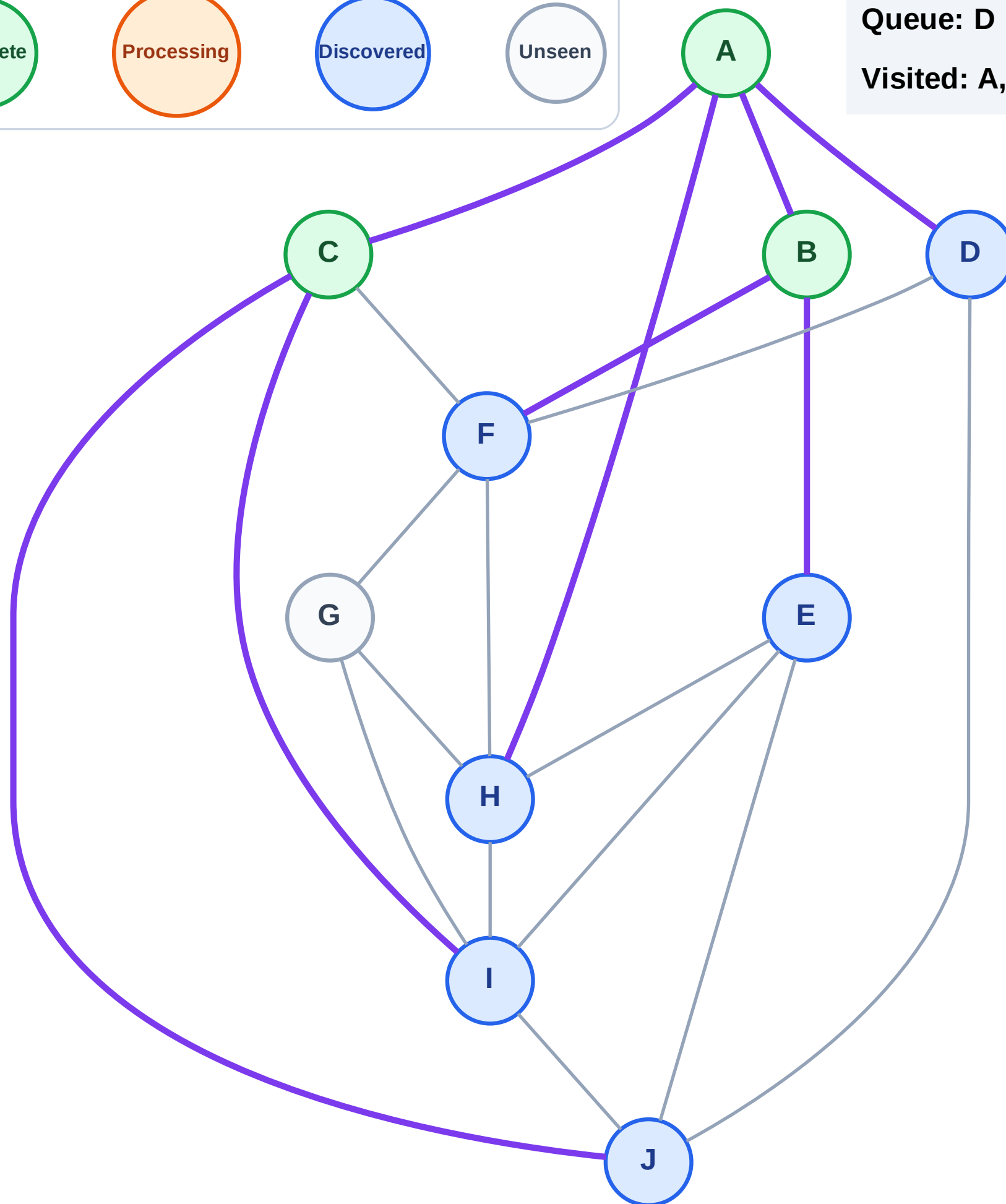


Discovered



Unseen

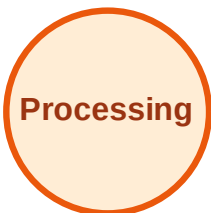
Queue: D → H → E → F → I → J
Visited: A, B, C



BFS Traversal

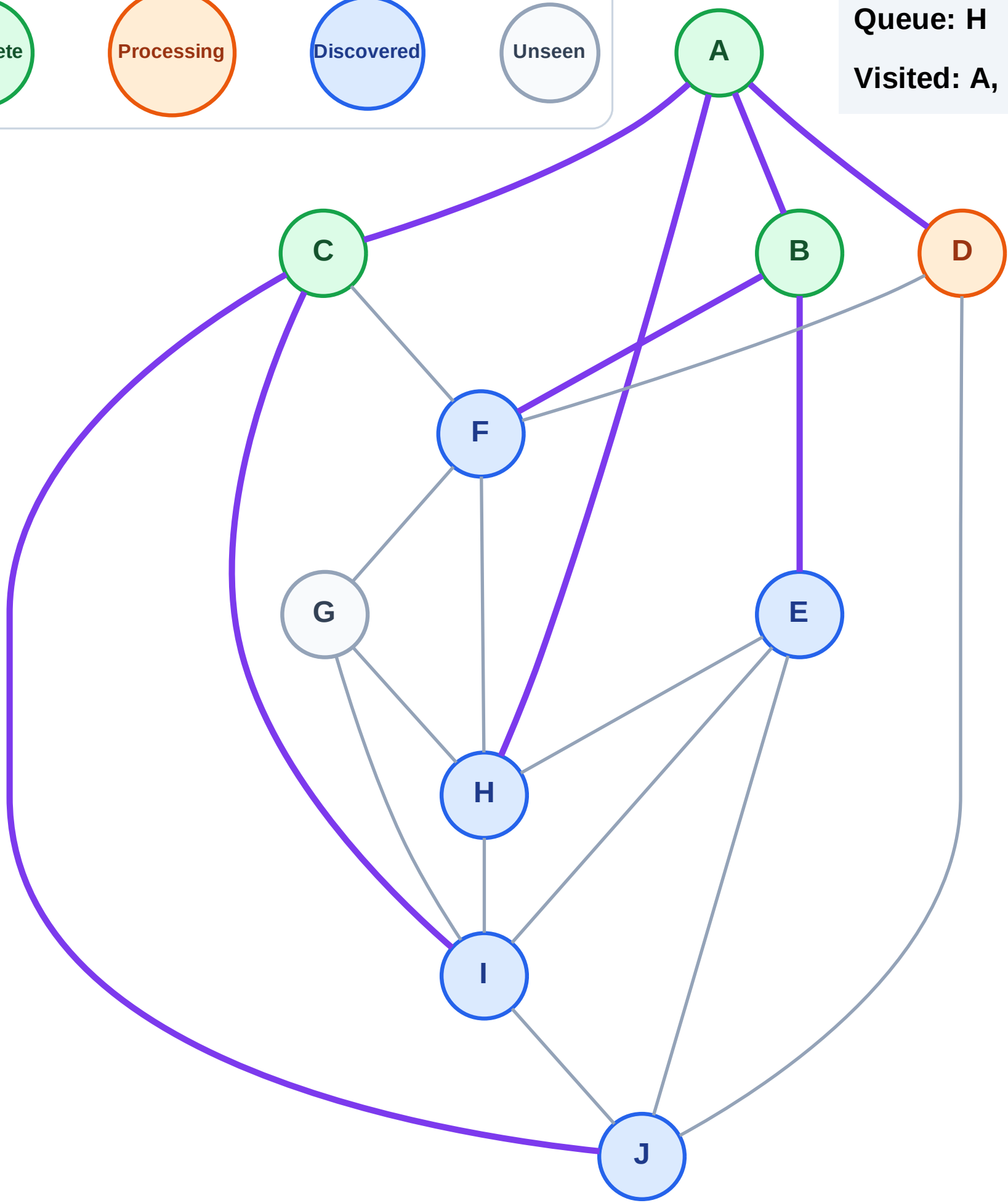
Step 8: Process node D

Legend



Queue: H → E → F → I → J

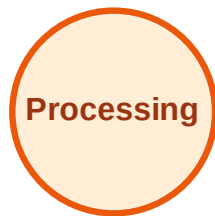
Visited: A, B, C



BFS Traversal

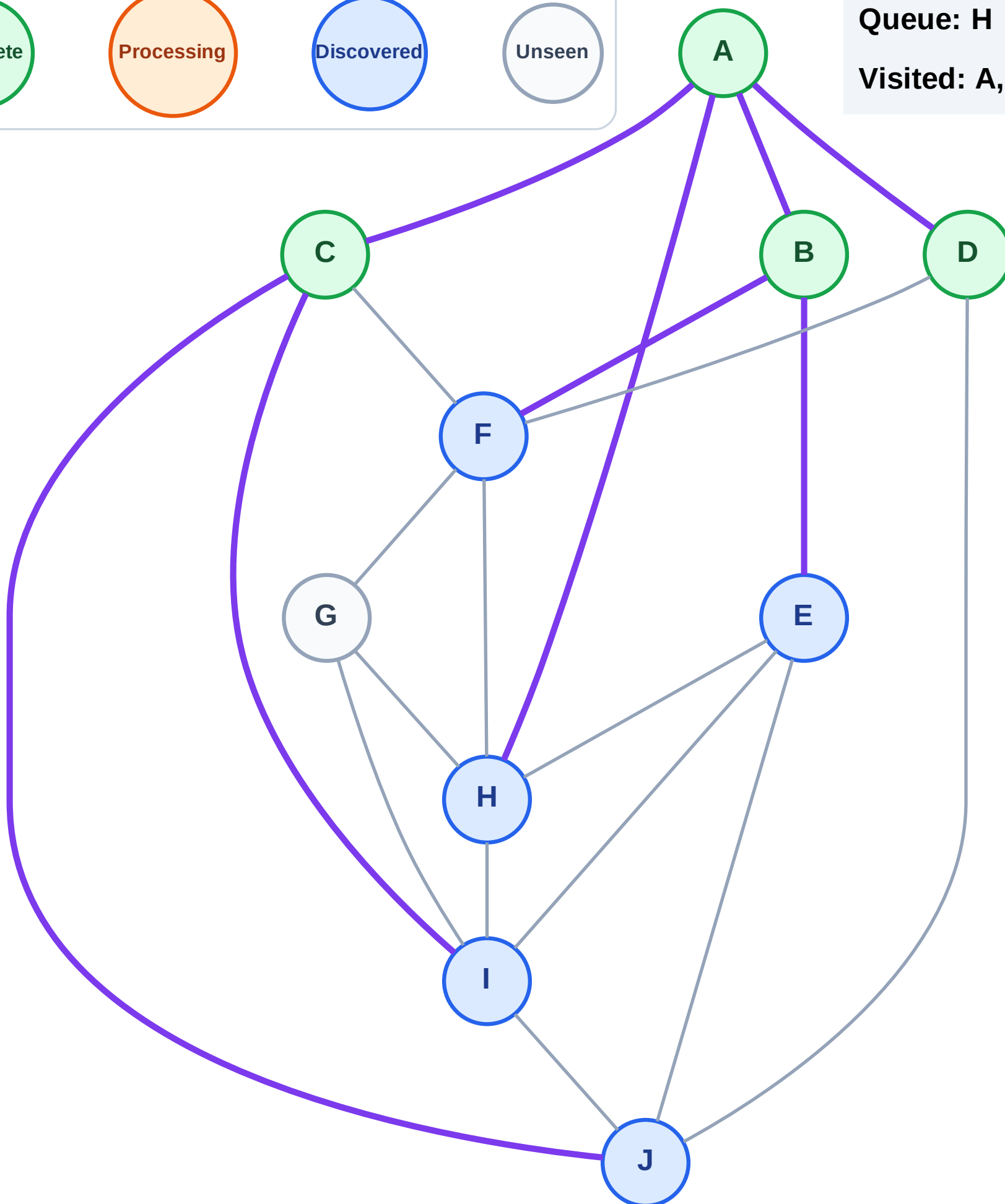
Step 9: No new neighbors from D

Legend



Queue: H → E → F → I → J

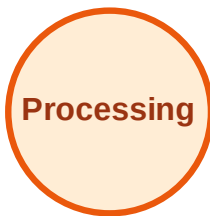
Visited: A, B, C, D



BFS Traversal

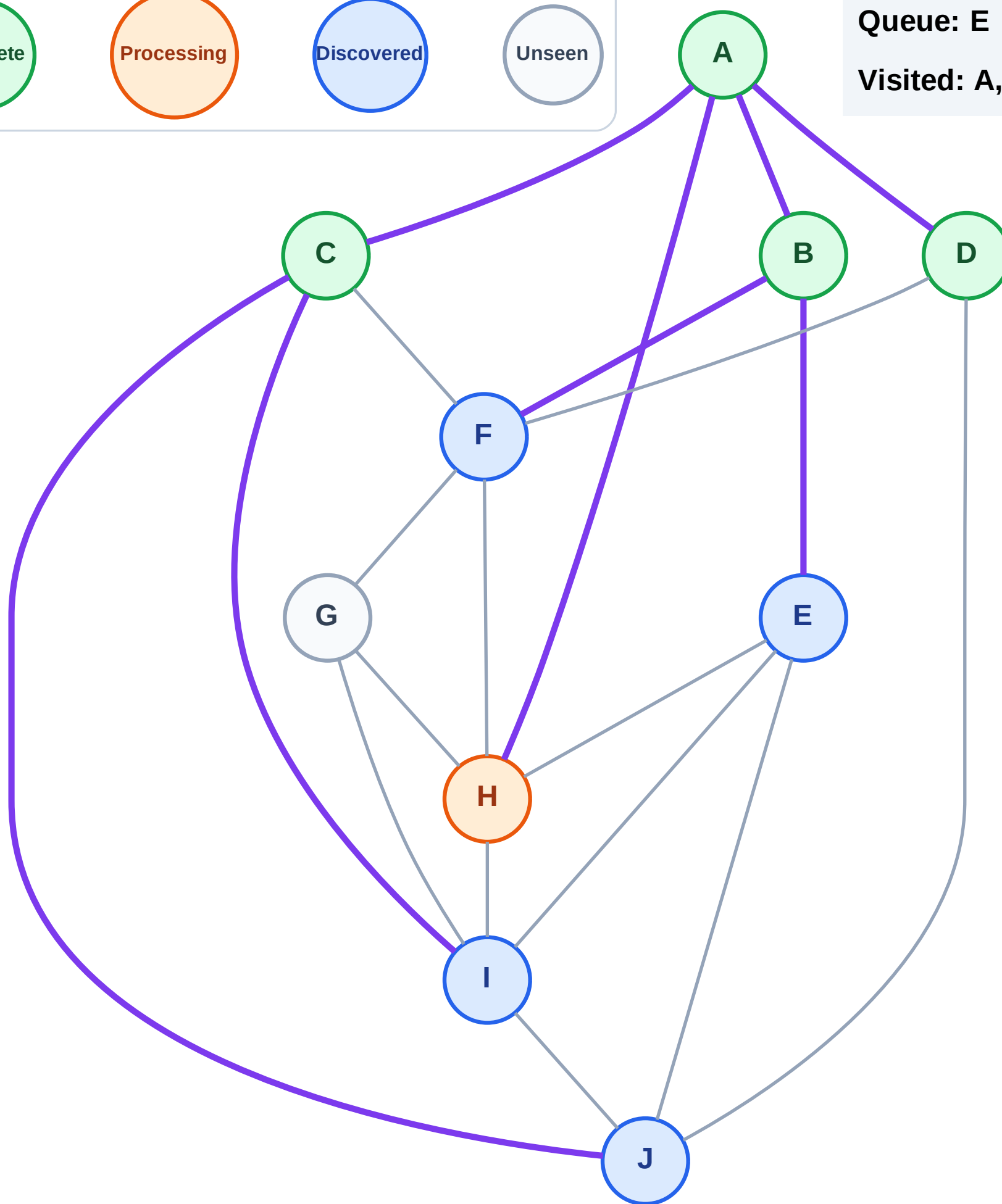
Step 10: Process node H

Legend



Queue: E → F → I → J

Visited: A, B, C, D



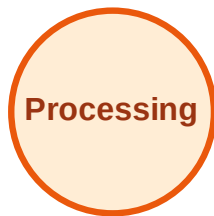
BFS Traversal

Step 11: Discover G from H

Legend



Complete



Processing



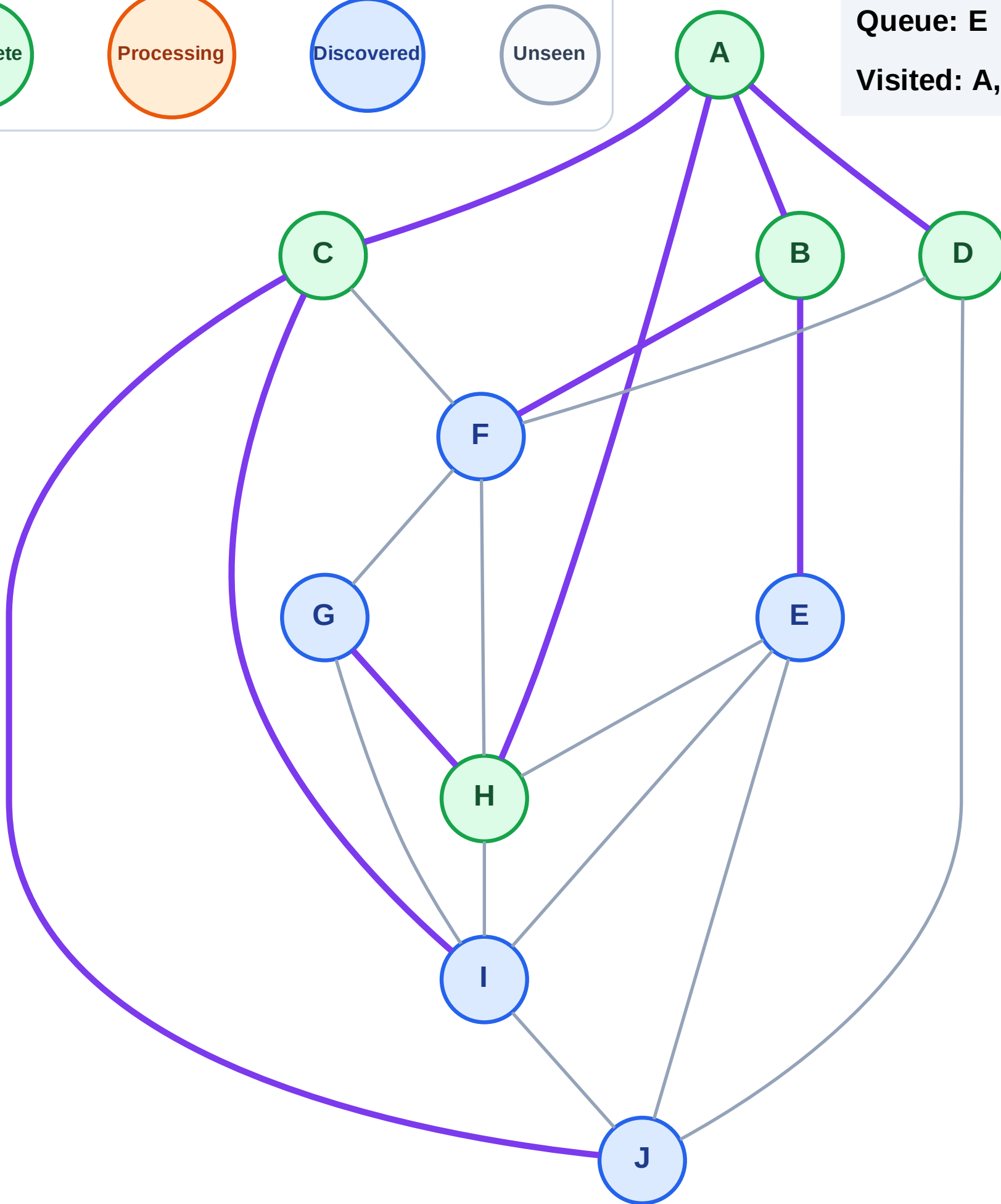
Discovered



Unseen

Queue: E → F → I → J → G

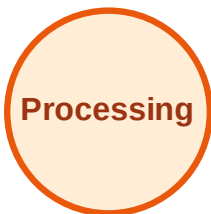
Visited: A, B, C, D, H



BFS Traversal

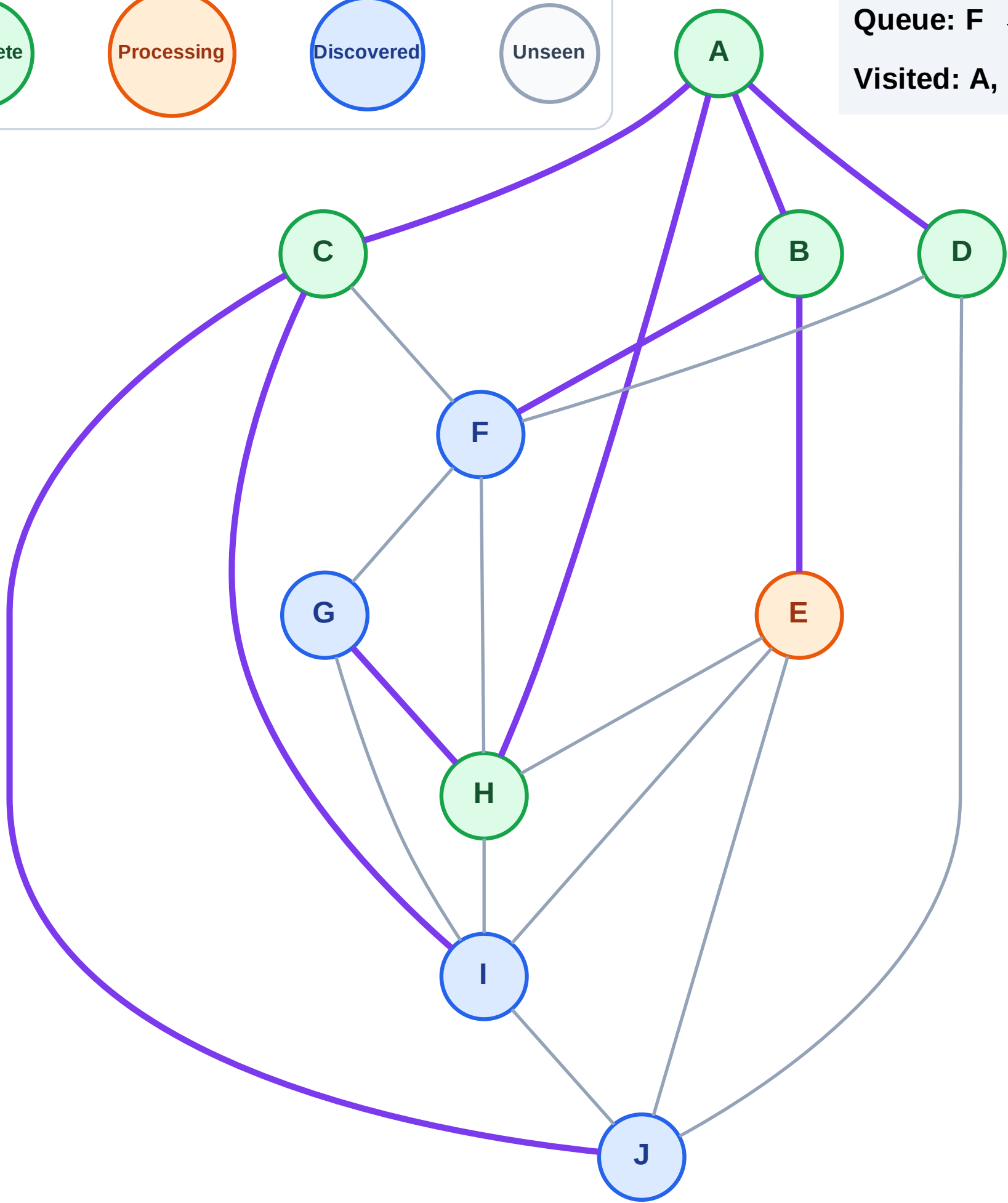
Step 12: Process node E

Legend



Queue: F → I → J → G

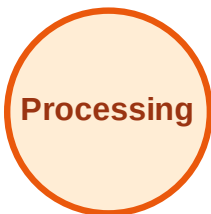
Visited: A, B, C, D, H



BFS Traversal

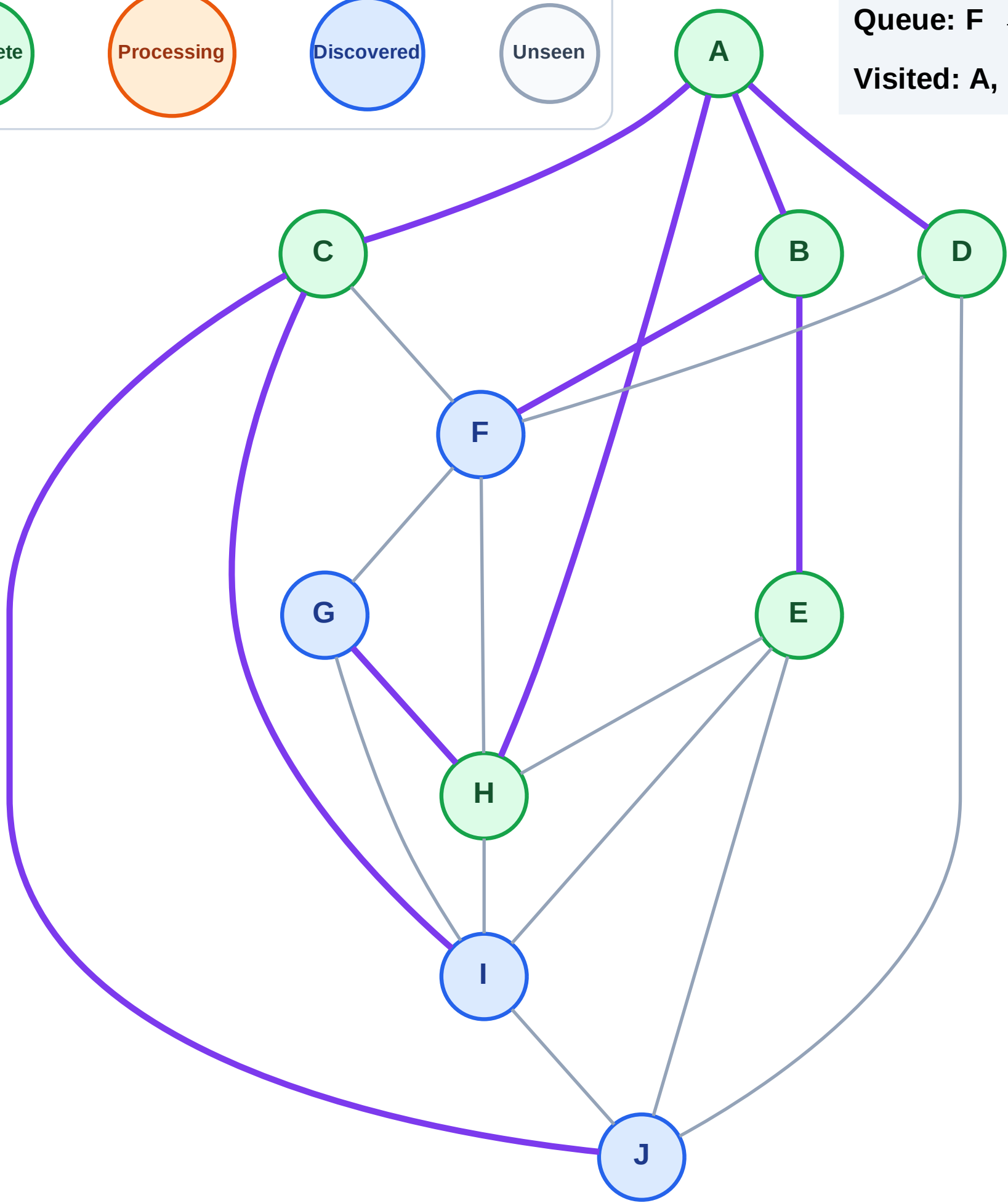
Step 13: No new neighbors from E

Legend



Queue: F → I → J → G

Visited: A, B, C, D, E, H



BFS Traversal

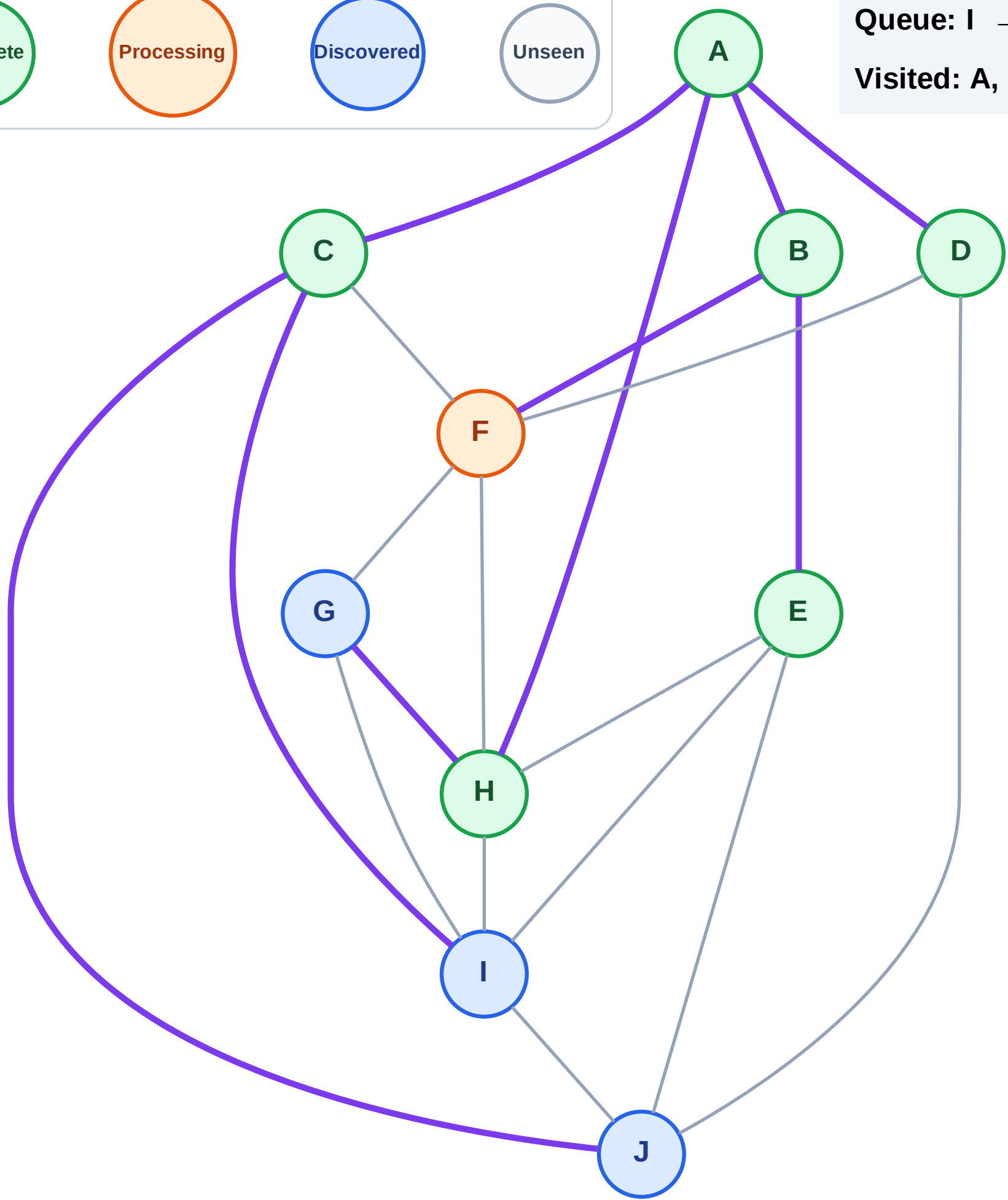
Step 14: Process node F

Legend



Queue: I → J → G

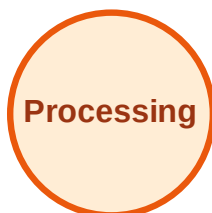
Visited: A, B, C, D, E, H



BFS Traversal

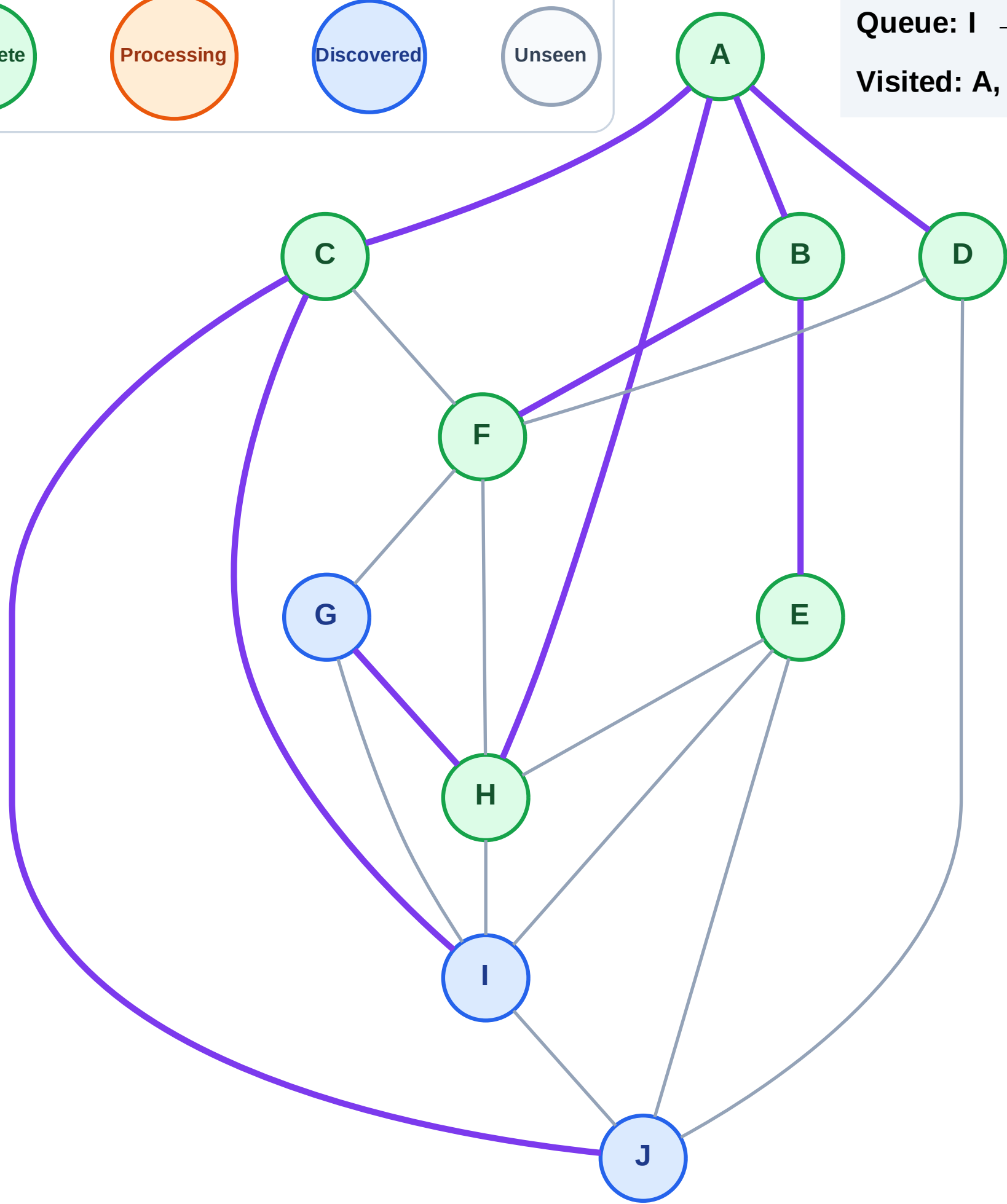
Step 15: No new neighbors from F

Legend



Queue: I → J → G

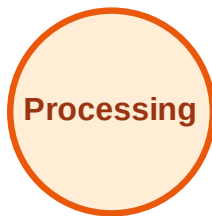
Visited: A, B, C, D, E, F, H



BFS Traversal

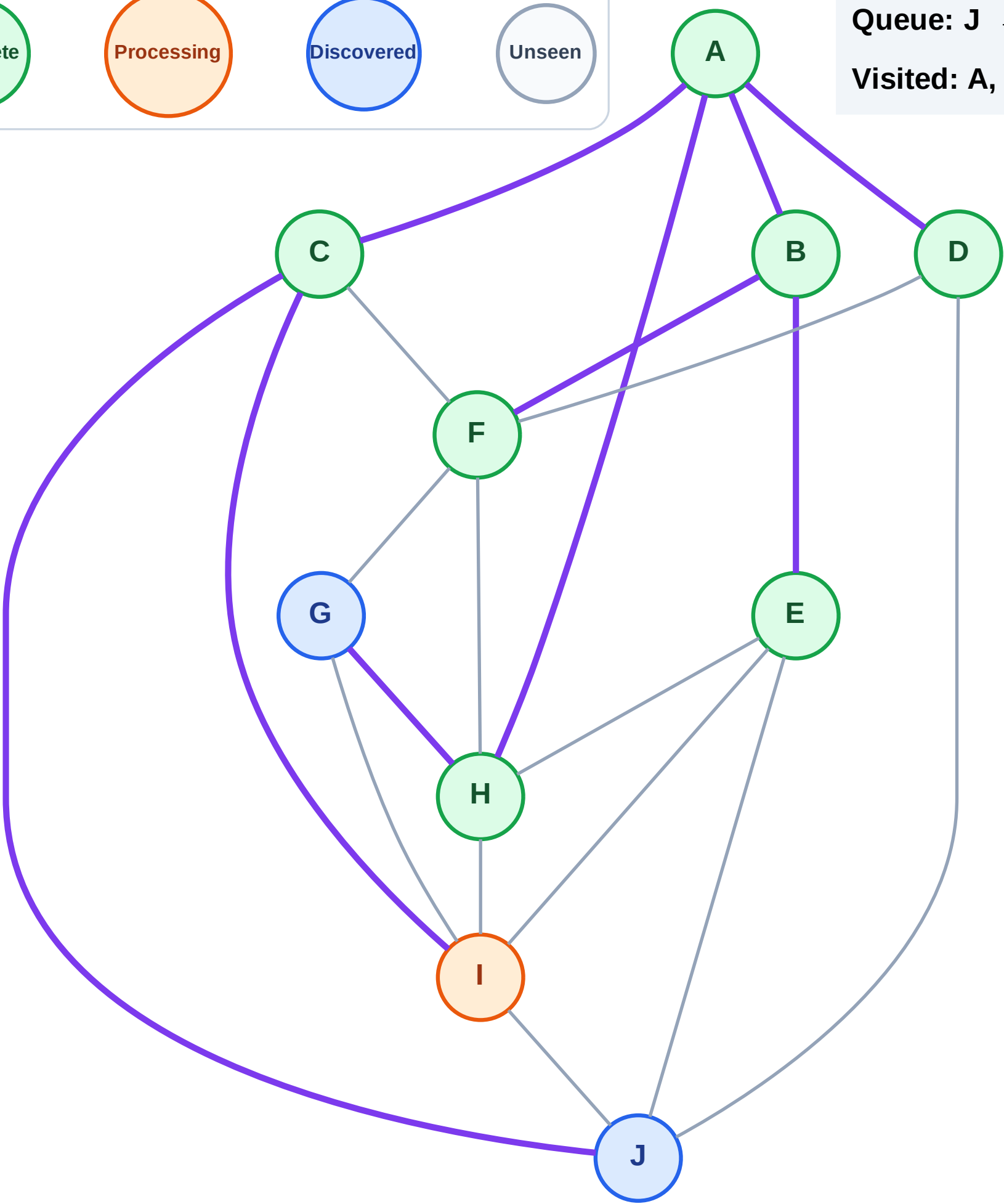
Step 16: Process node I

Legend



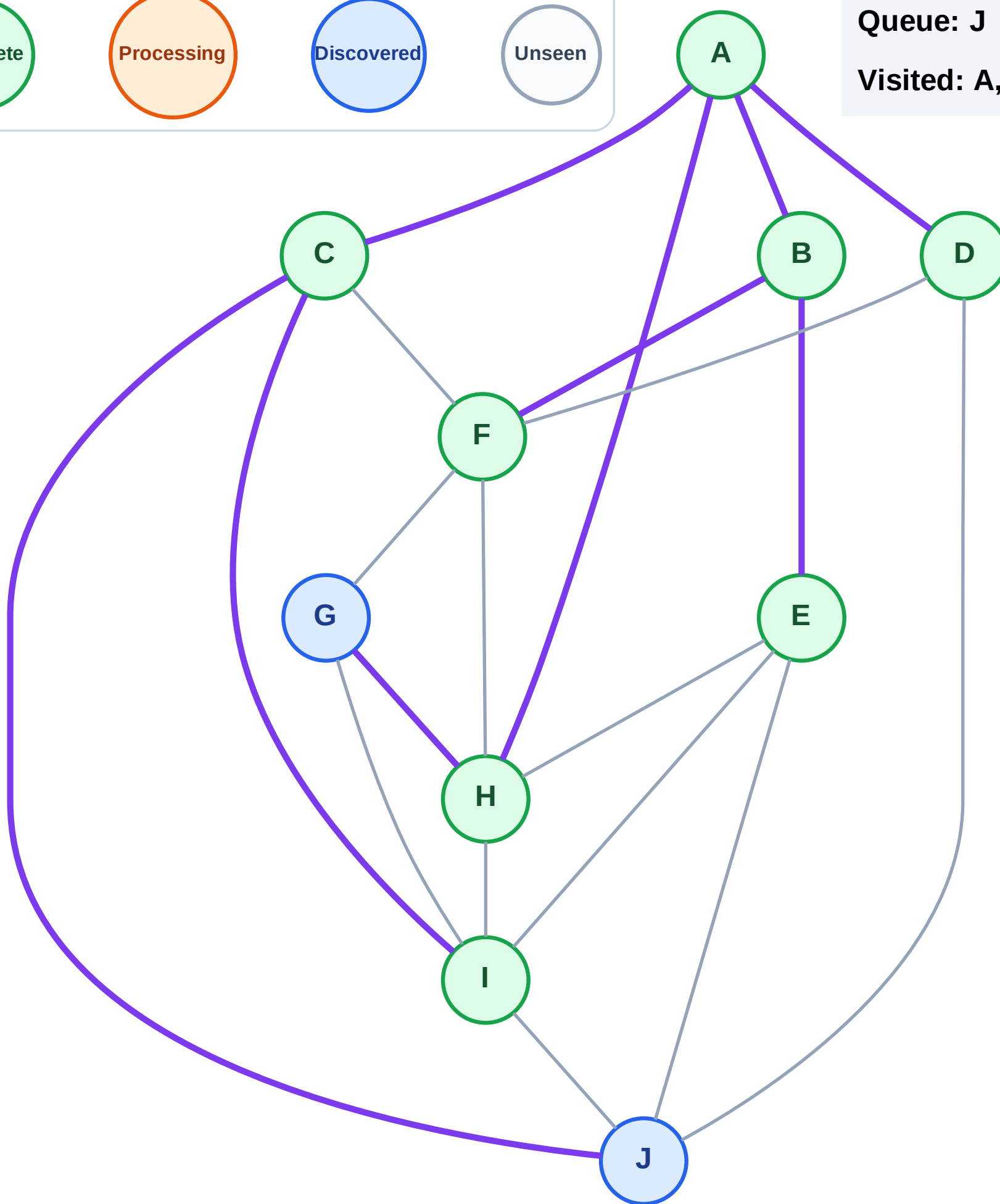
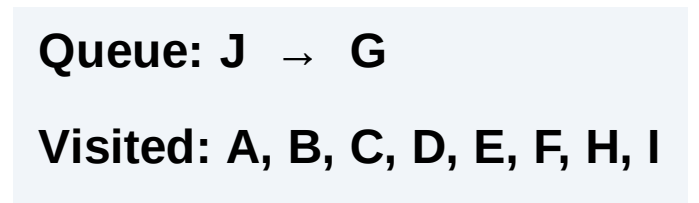
Queue: J → G

Visited: A, B, C, D, E, F, H



Step 17: No new neighbors from I

Step 17: No new neighbors from I



BFS Traversal

Step 18: Process node J

Legend

Complete

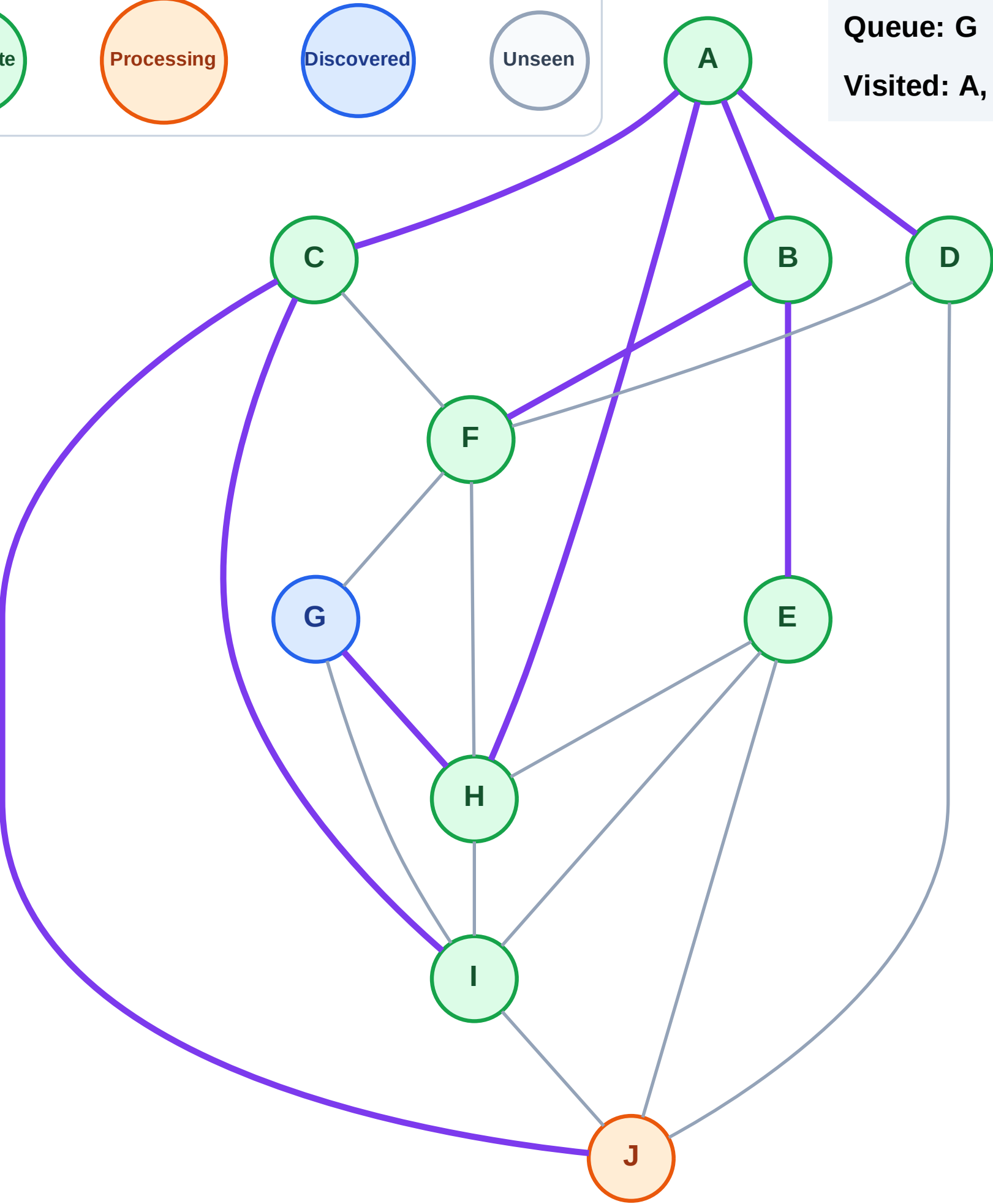
Processing

Discovered

Unseen

Queue: G

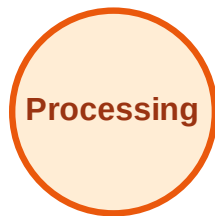
Visited: A, B, C, D, E, F, H, I



BFS Traversal

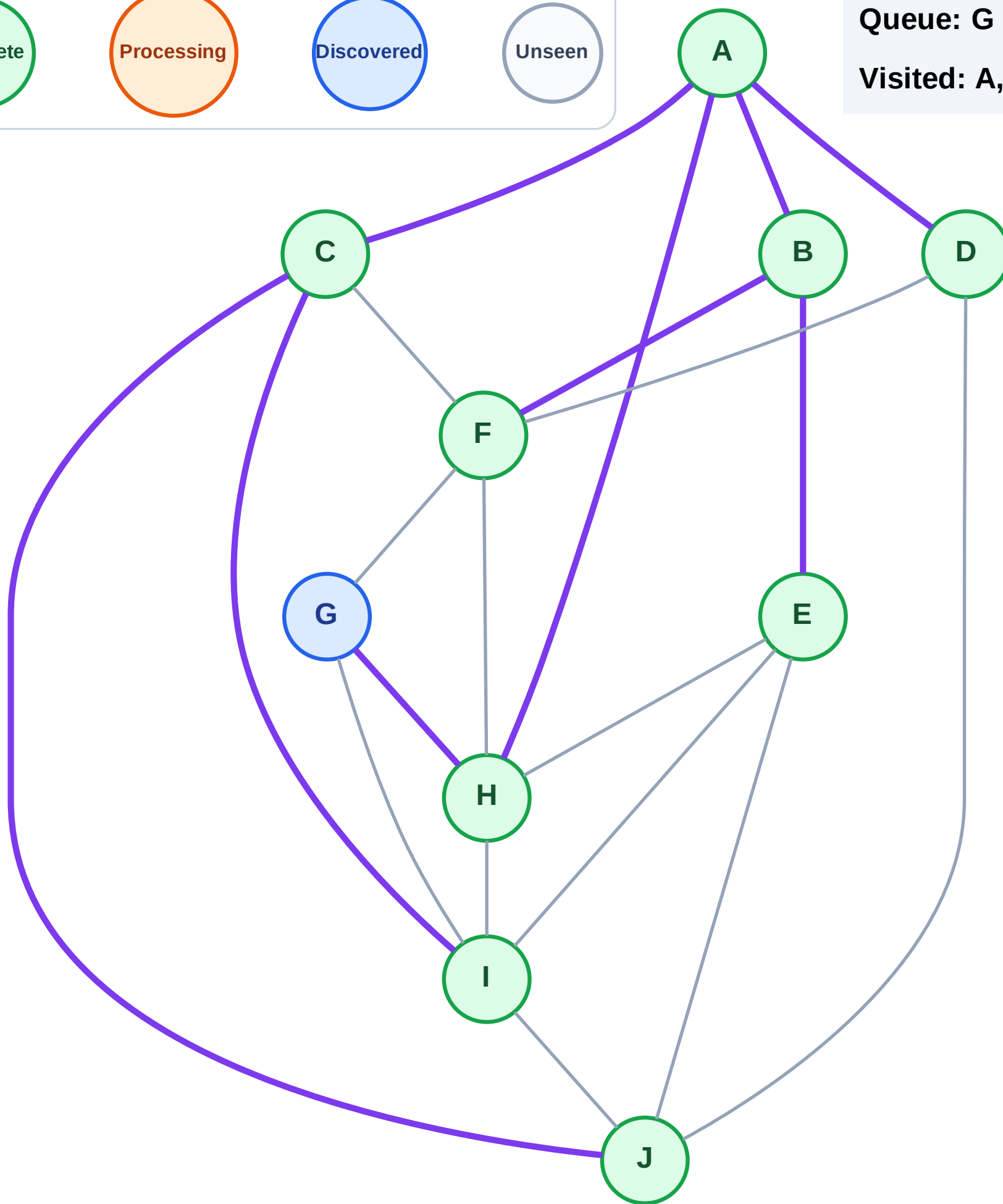
Step 19: No new neighbors from J

Legend



Queue: G

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

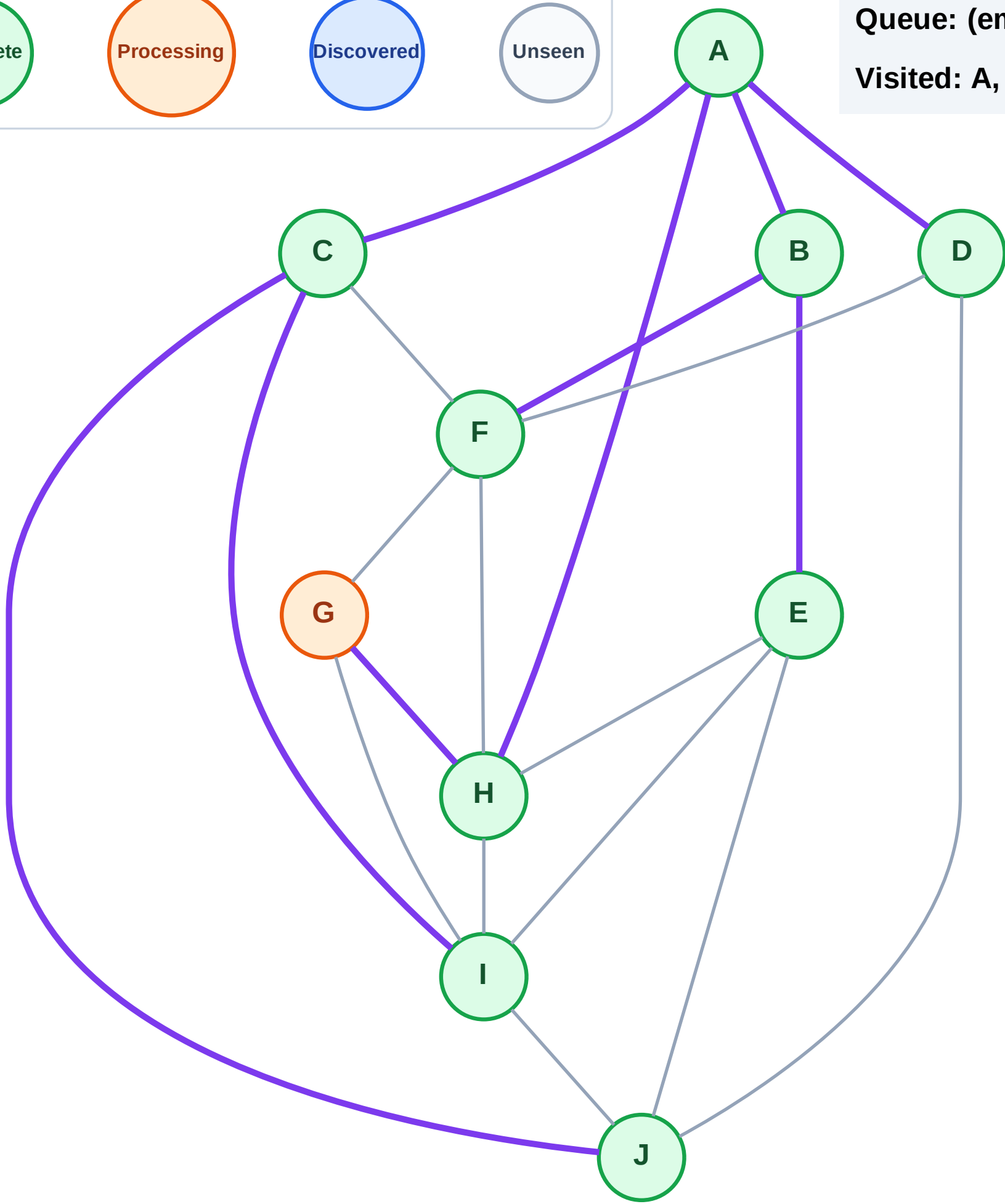
Step 20: Process node G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J



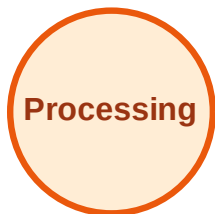
BFS Traversal

Step 21: No new neighbors from G

Legend



Complete



Processing



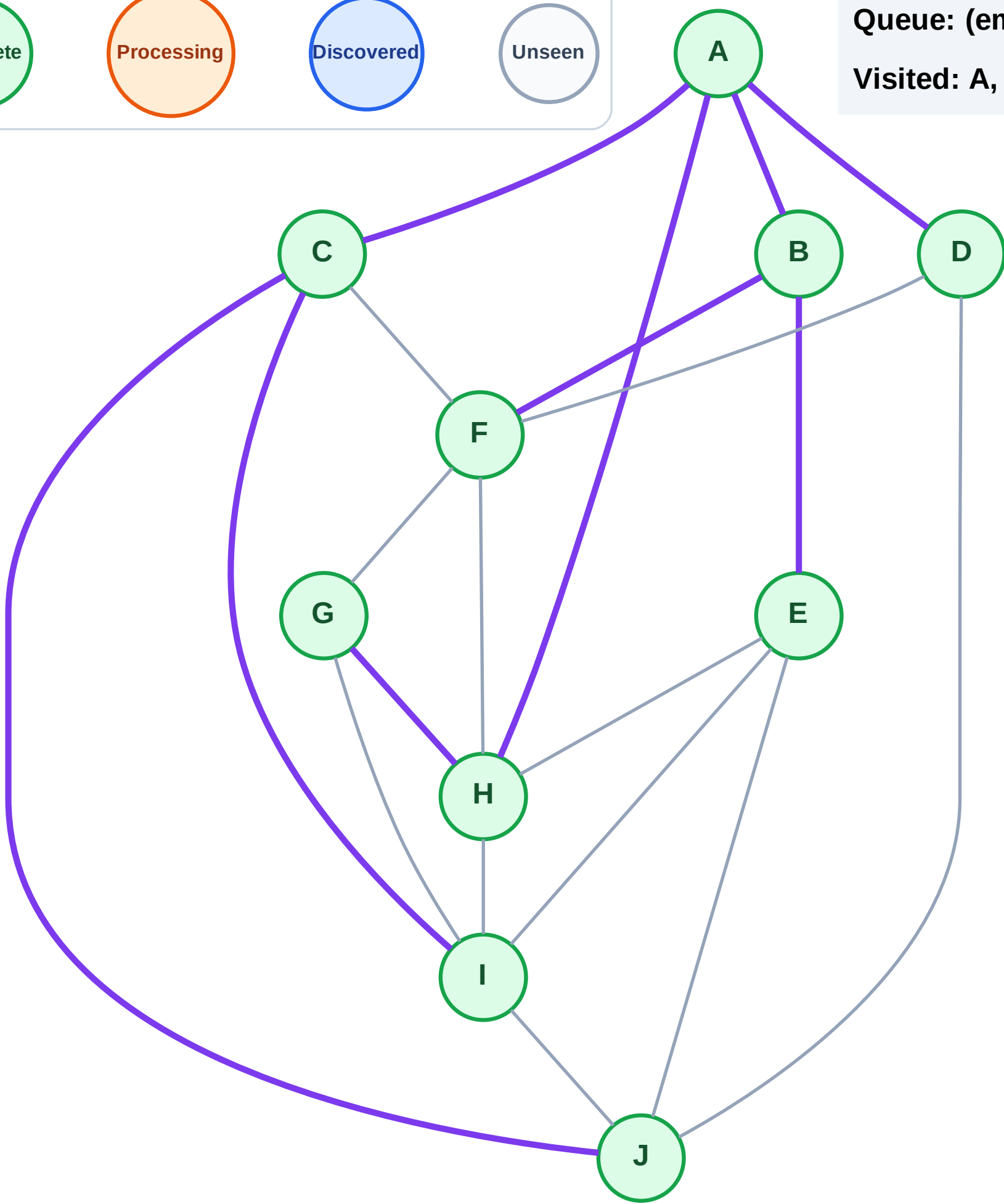
Discovered



Unseen

Queue: (empty)

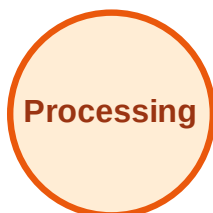
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

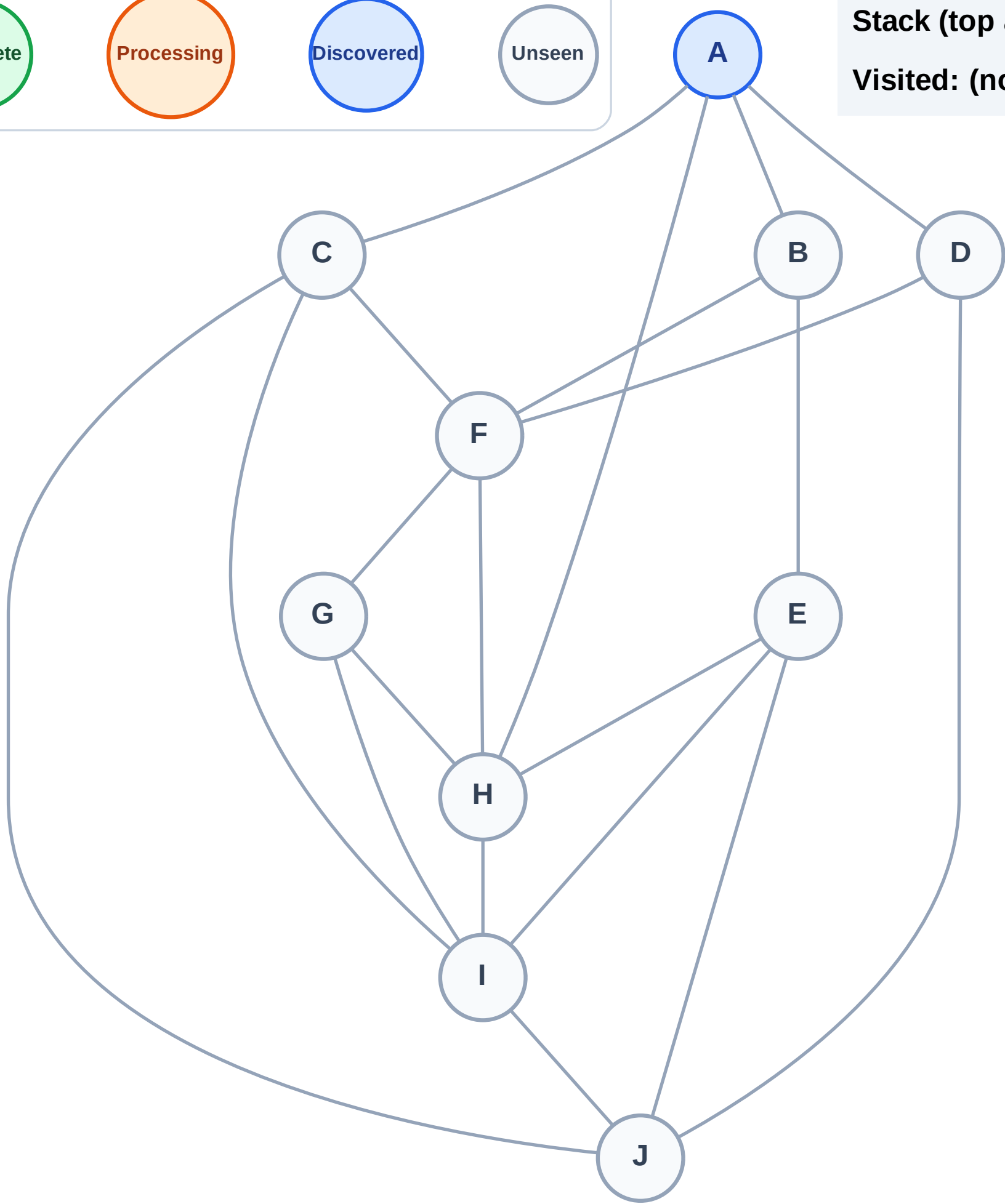
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

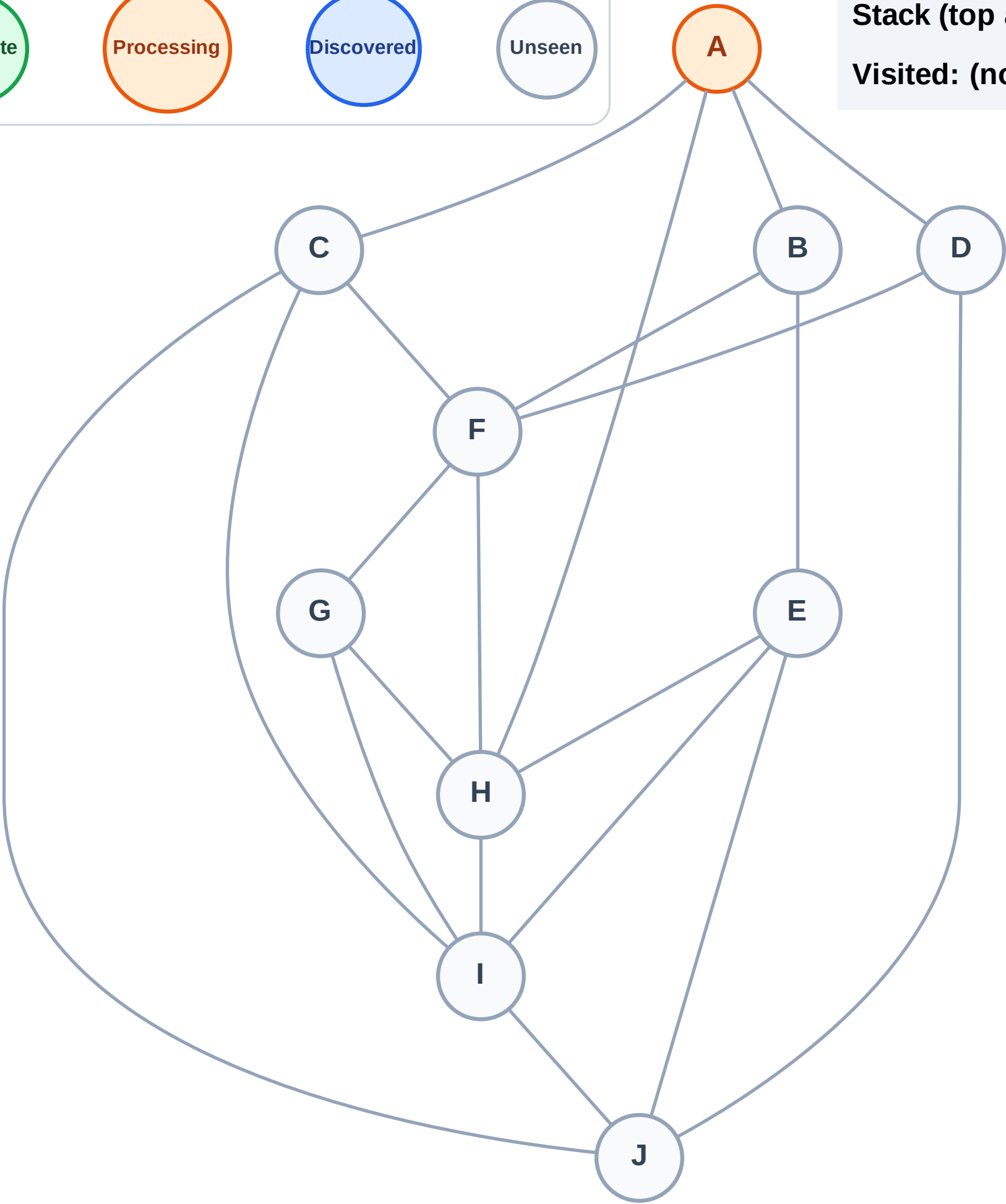
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover H, D, C, B from A

Legend

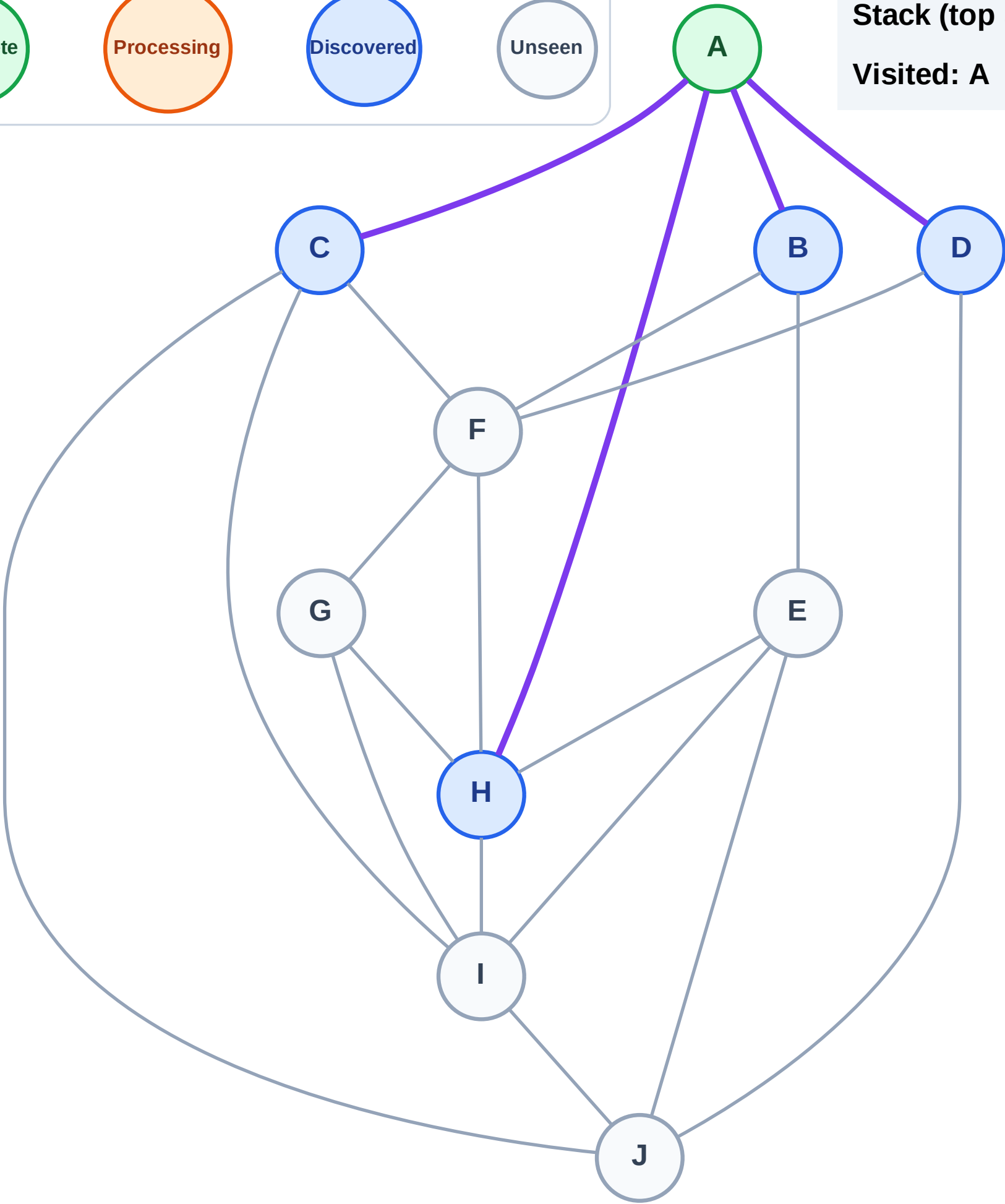
Complete

Processing

Discovered

Unseen

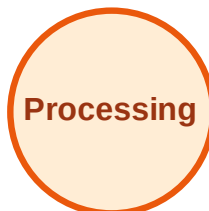
Stack (top at right): H | D | C | B
Visited: A



DFS Traversal

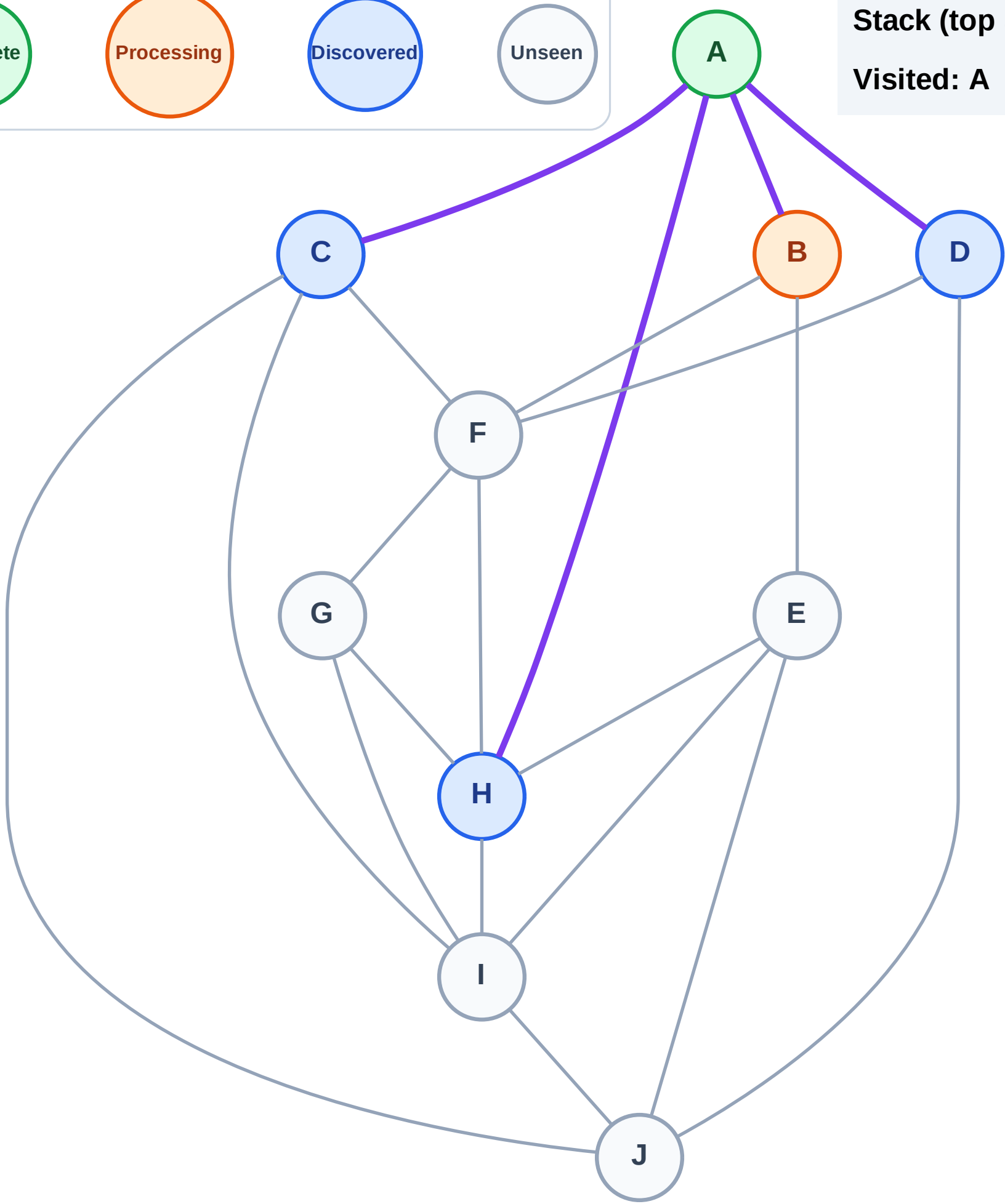
Step 4: Process node B

Legend



Stack (top at right): H | D | C

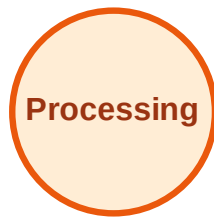
Visited: A



DFS Traversal

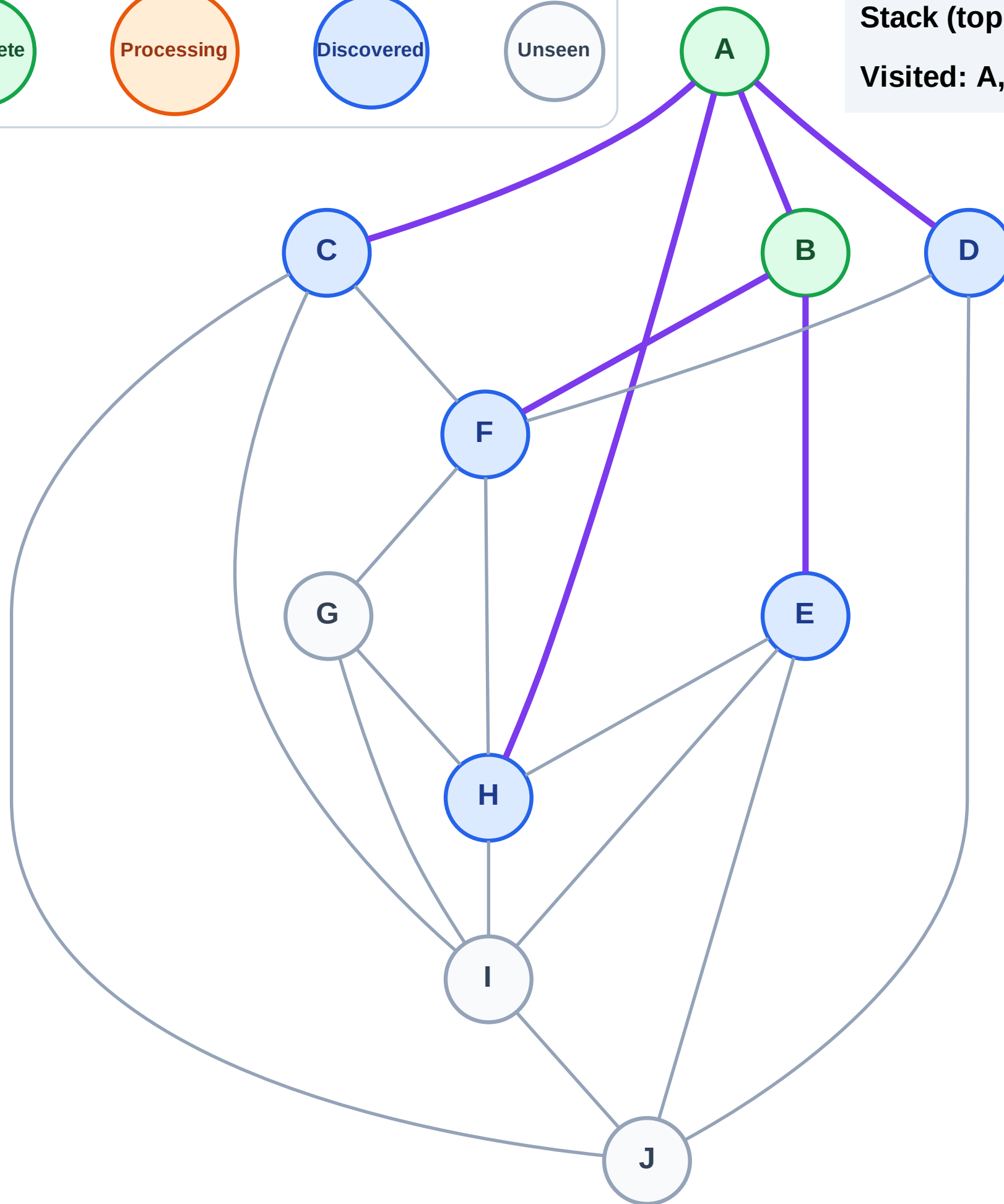
Step 5: Discover F, E from B

Legend



Stack (top at right): H | D | C | F | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

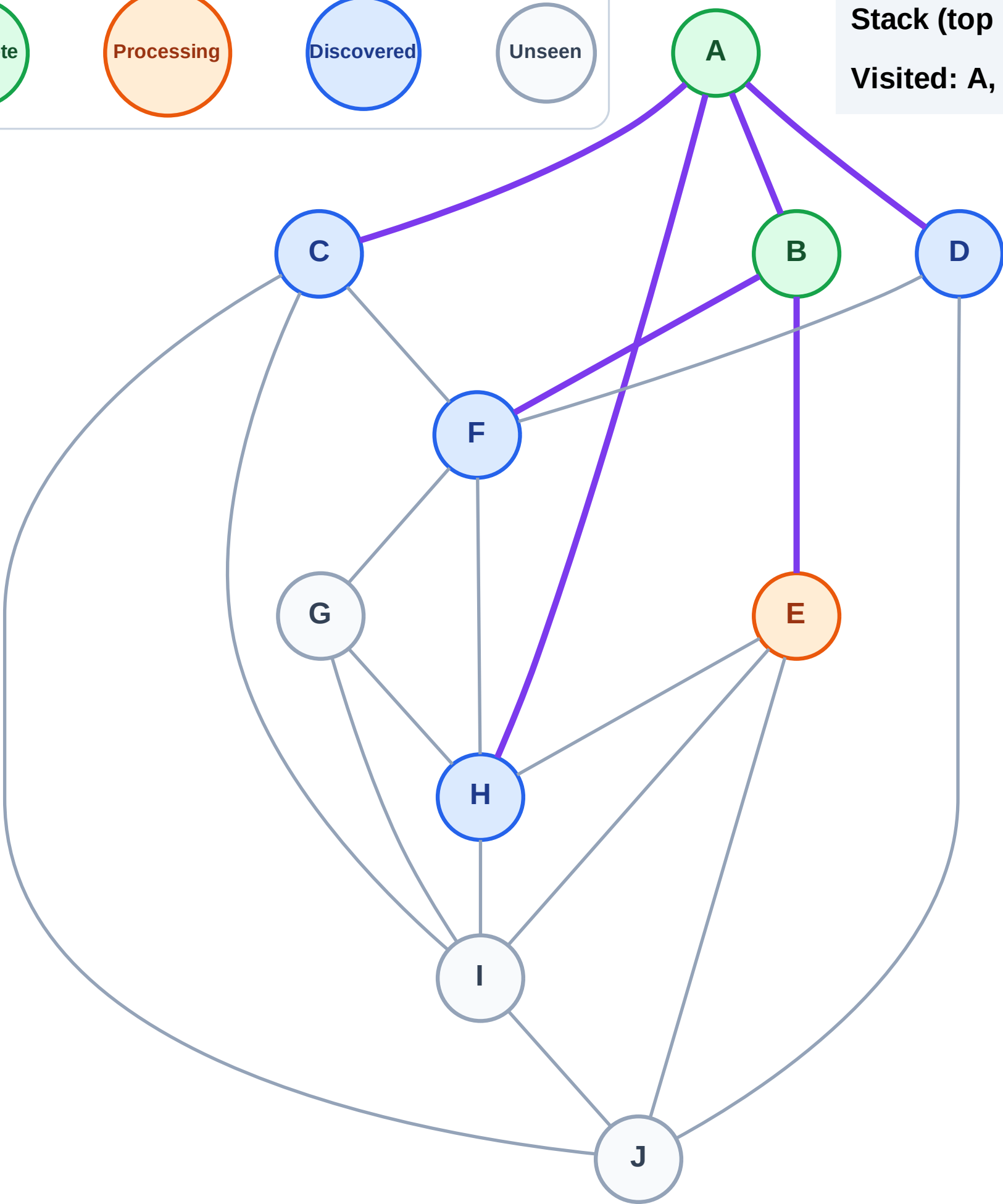
Complete

Processing

Discovered

Unseen

Stack (top at right): H | D | C | F
Visited: A, B



DFS Traversal

Step 7: Discover J, I from E

Legend



Complete



Processing



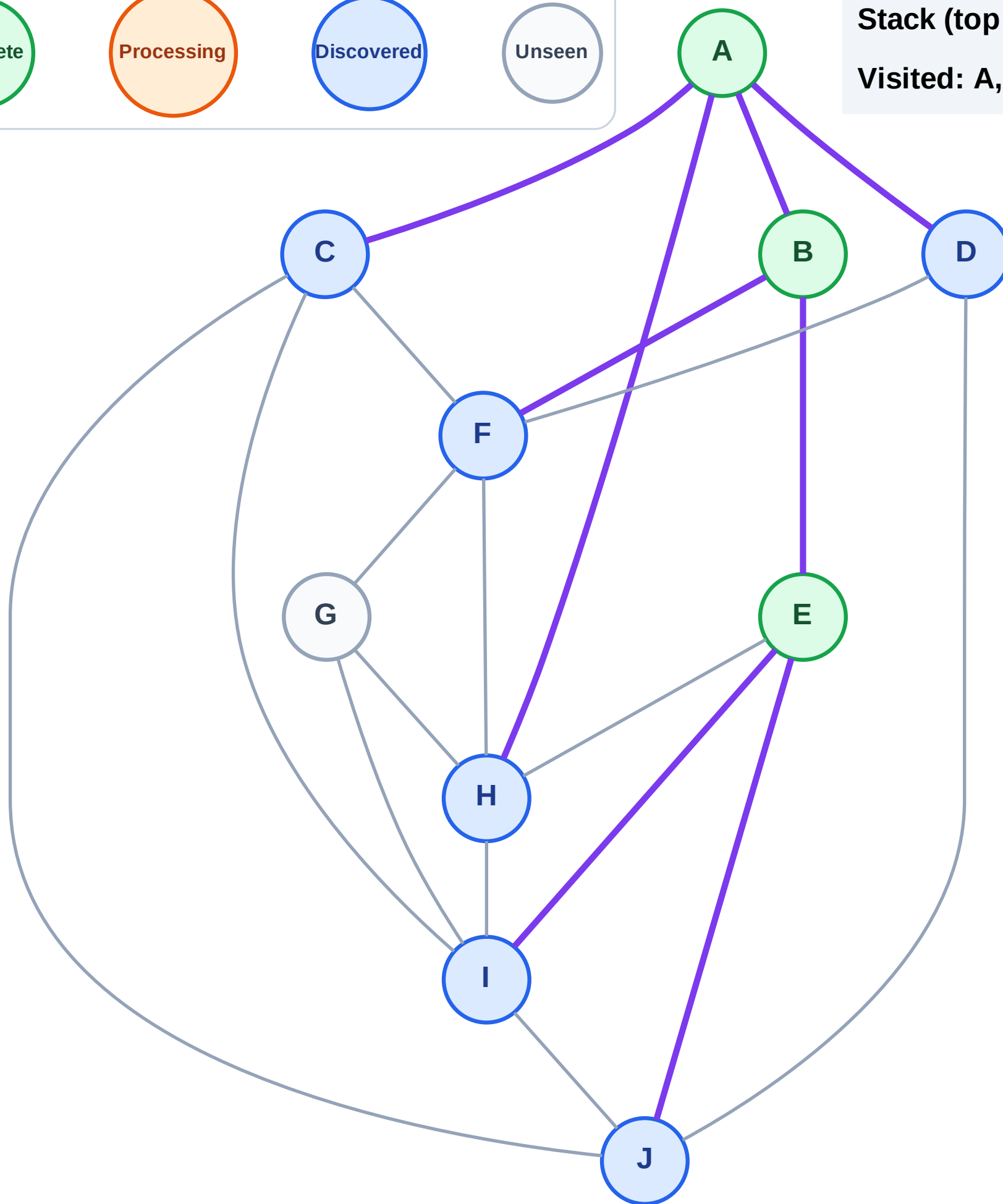
Discovered



Unseen

Stack (top at right): H | D | C | F | J | I

Visited: A, B, E



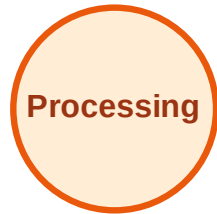
DFS Traversal

Step 8: Process node I

Legend



Complete



Processing



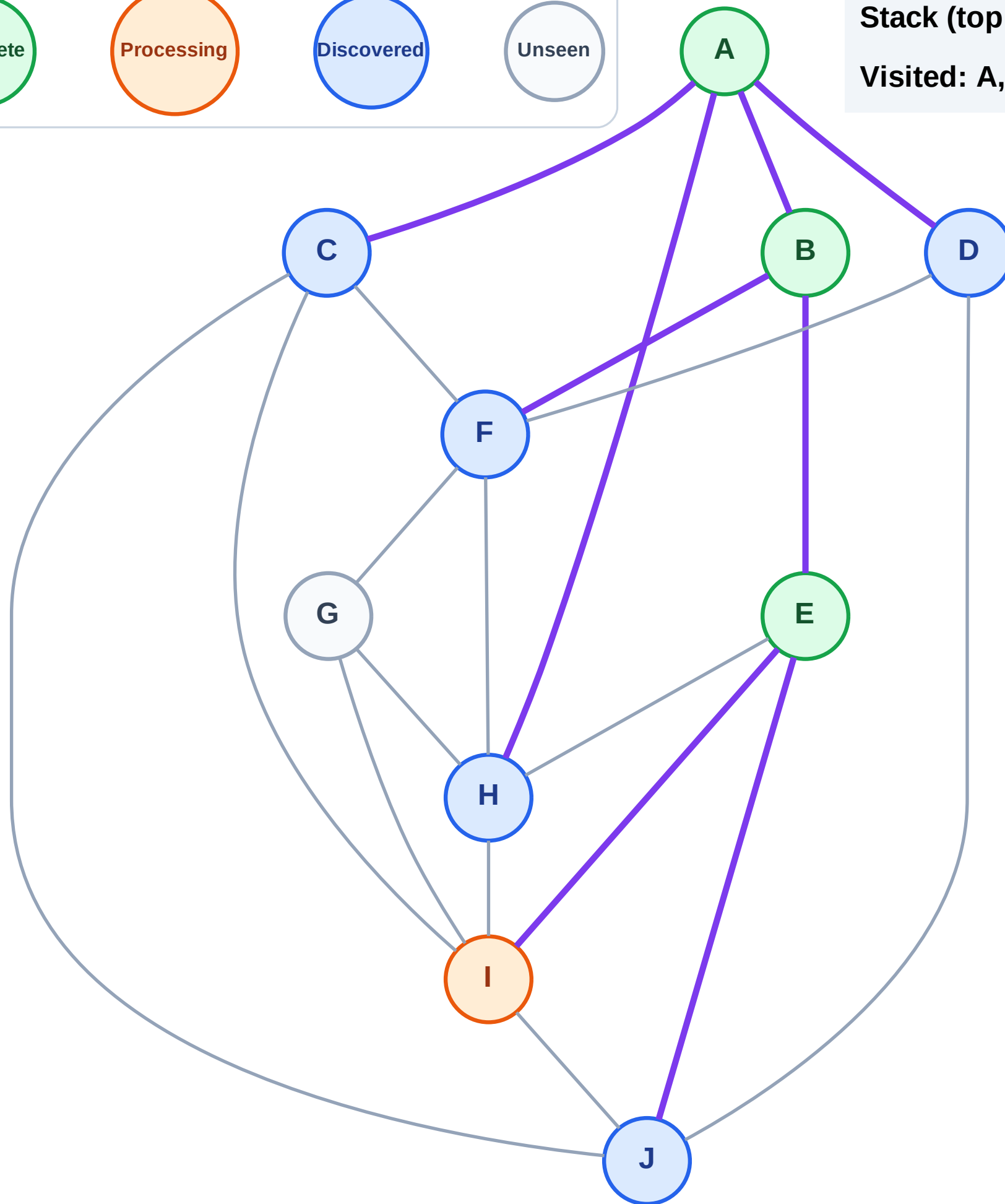
Discovered



Unseen

Stack (top at right): H | D | C | F | J

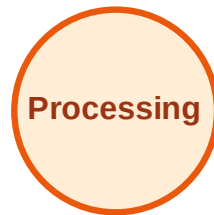
Visited: A, B, E



DFS Traversal

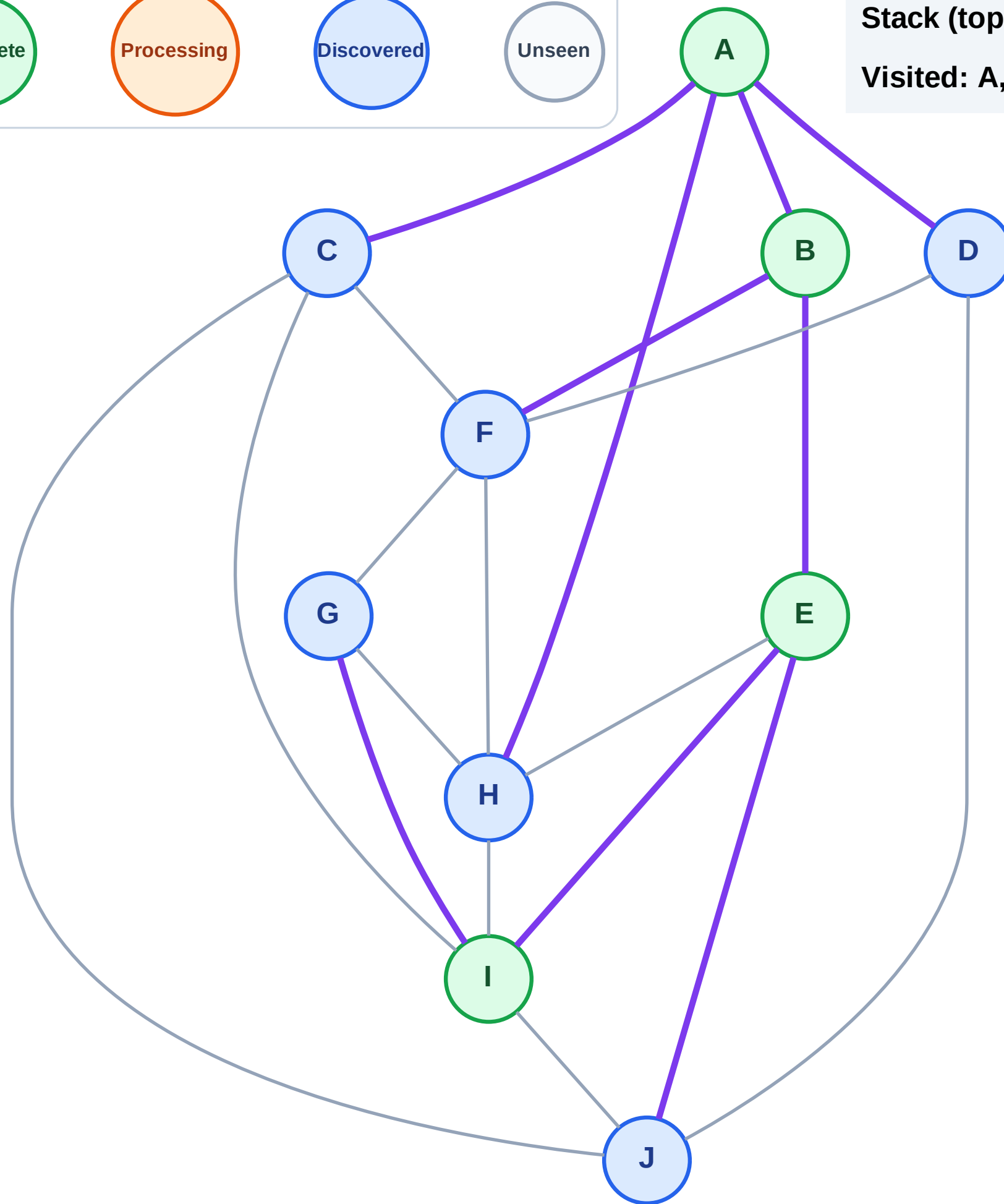
Step 9: Discover G from I

Legend



Stack (top at right): H | D | C | F | J | G

Visited: A, B, E, I



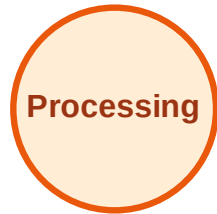
DFS Traversal

Step 10: Process node G

Legend



Complete



Processing



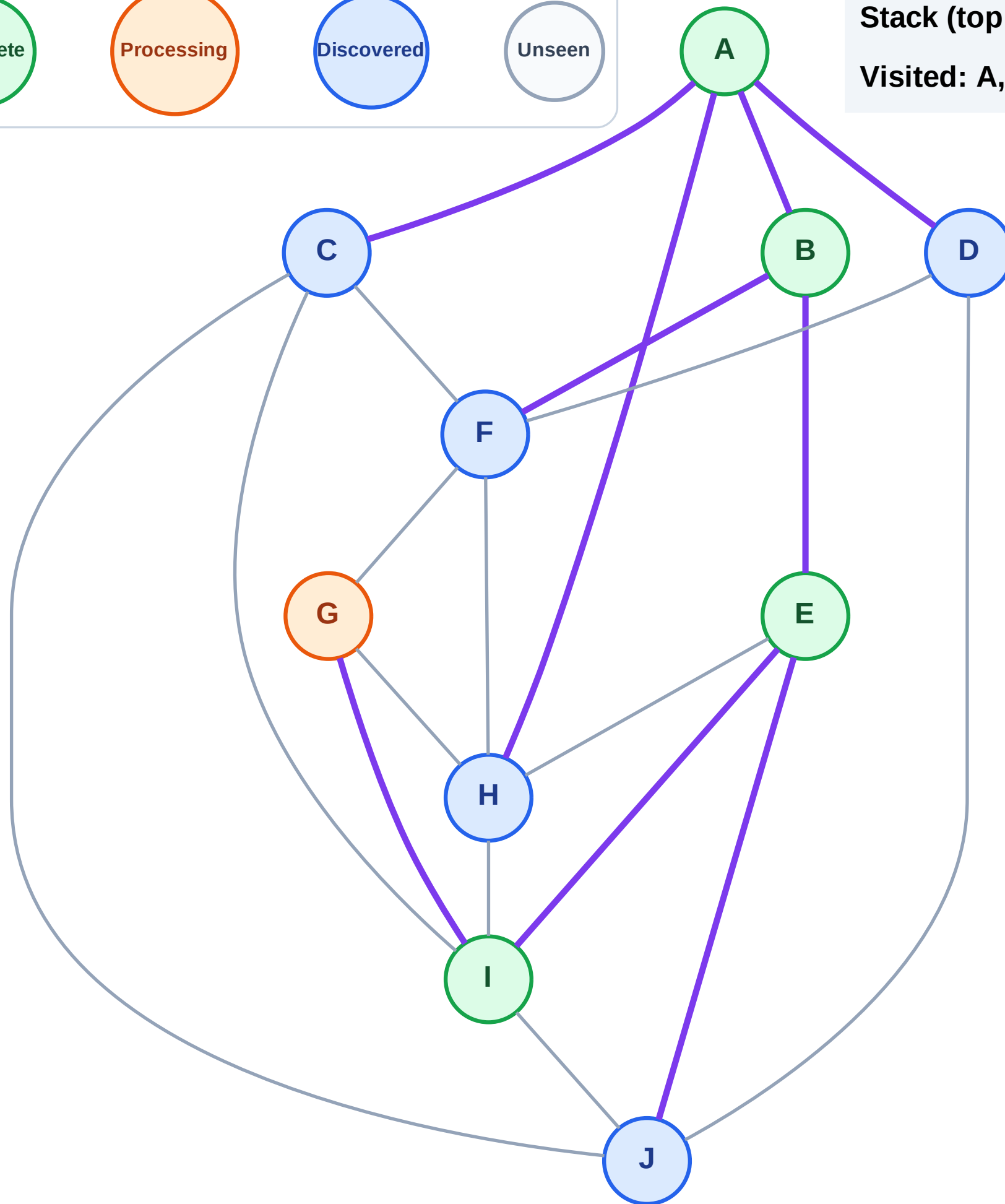
Discovered



Unseen

Stack (top at right): H | D | C | F | J

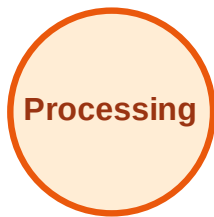
Visited: A, B, E, I



DFS Traversal

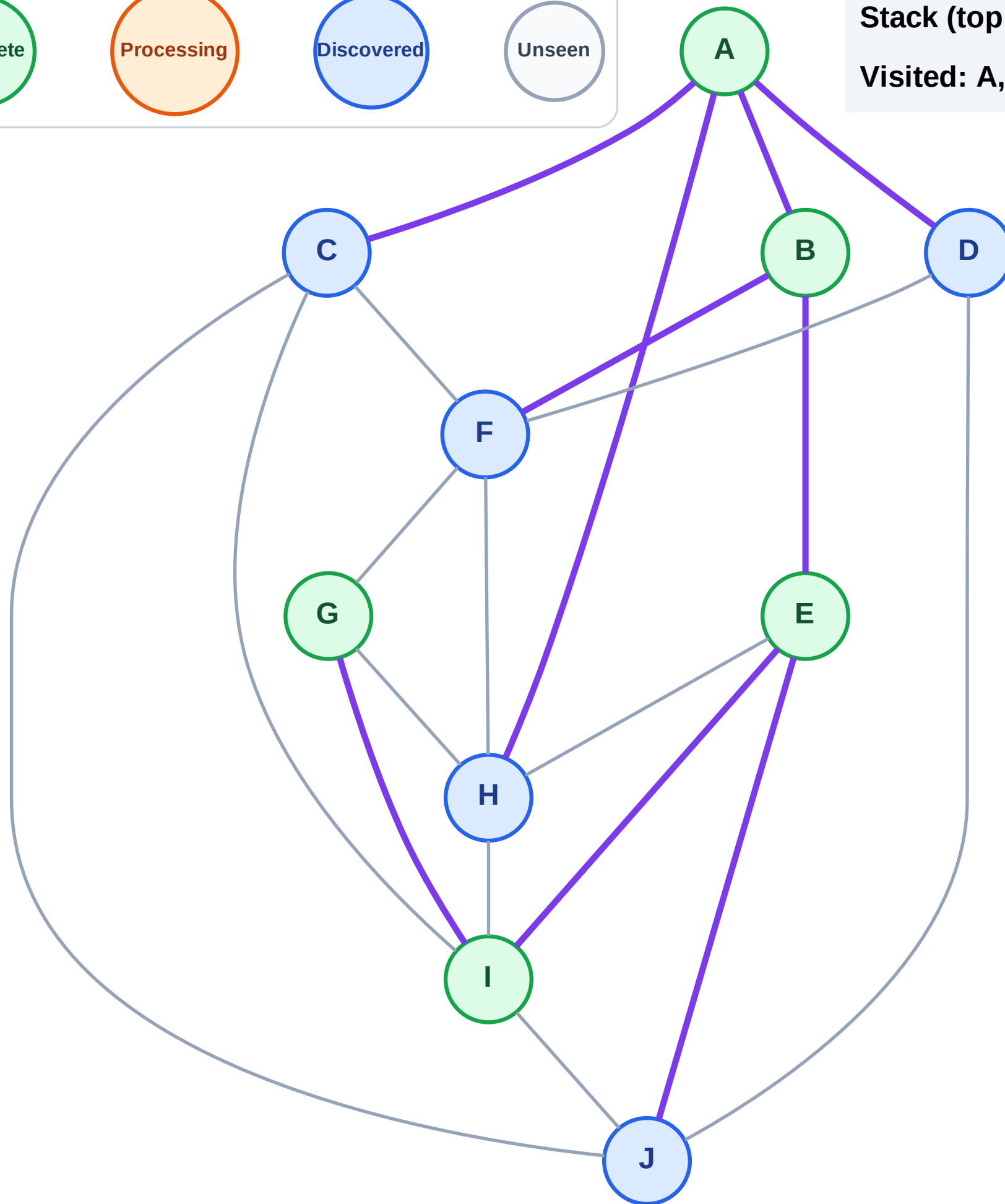
Step 11: No new neighbors from G

Legend



Stack (top at right): H | D | C | F | J

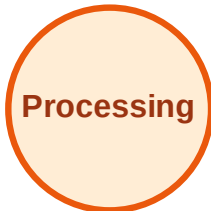
Visited: A, B, E, G, I



DFS Traversal

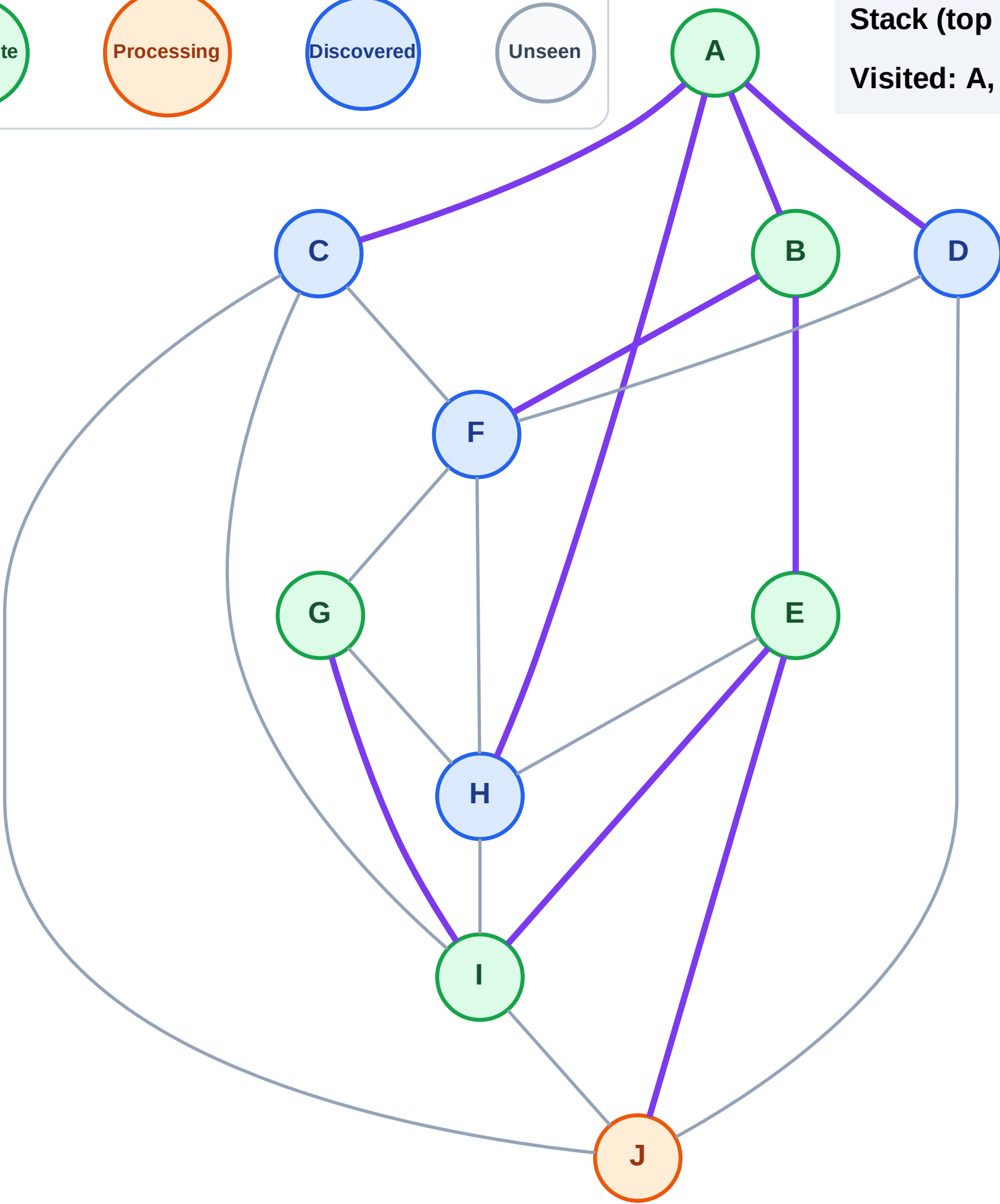
Step 12: Process node J

Legend



Stack (top at right): H | D | C | F

Visited: A, B, E, G, I



DFS Traversal

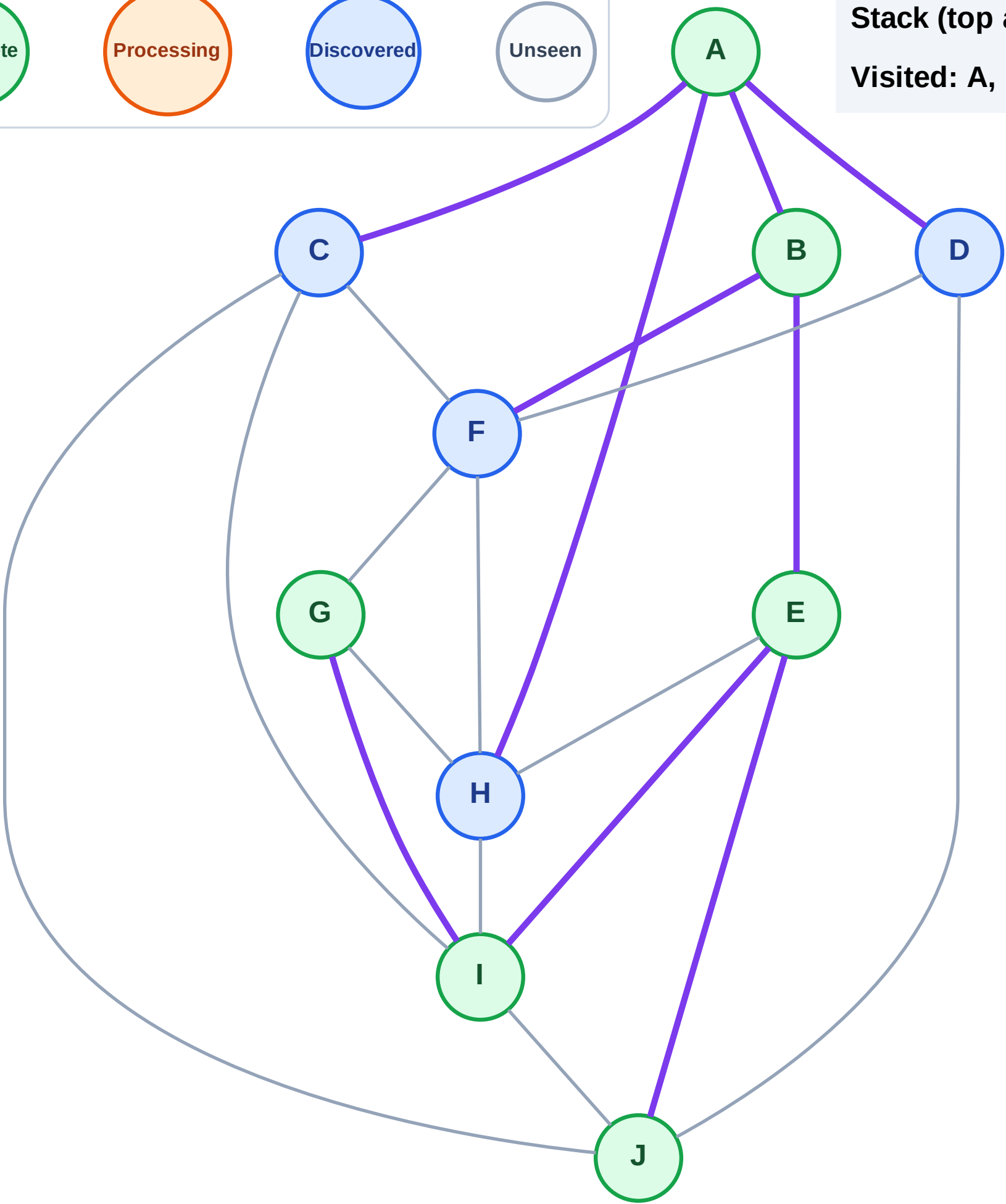
Step 13: No new neighbors from J

Legend



Stack (top at right): H | D | C | F

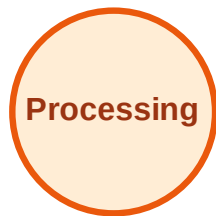
Visited: A, B, E, G, I, J



DFS Traversal

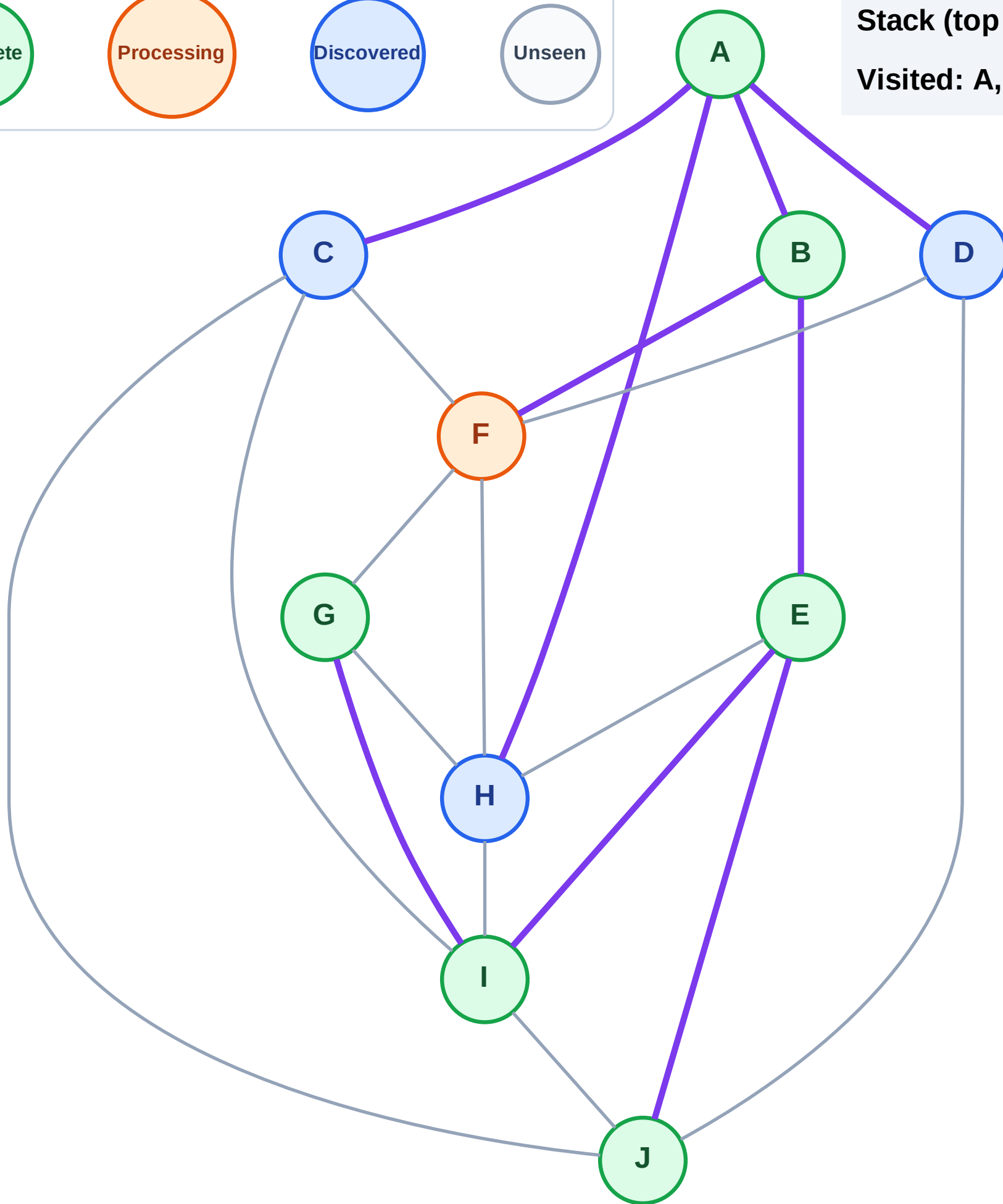
Step 14: Process node F

Legend



Stack (top at right): H | D | C

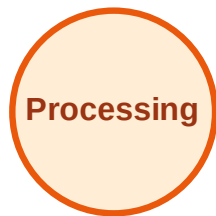
Visited: A, B, E, G, I, J



DFS Traversal

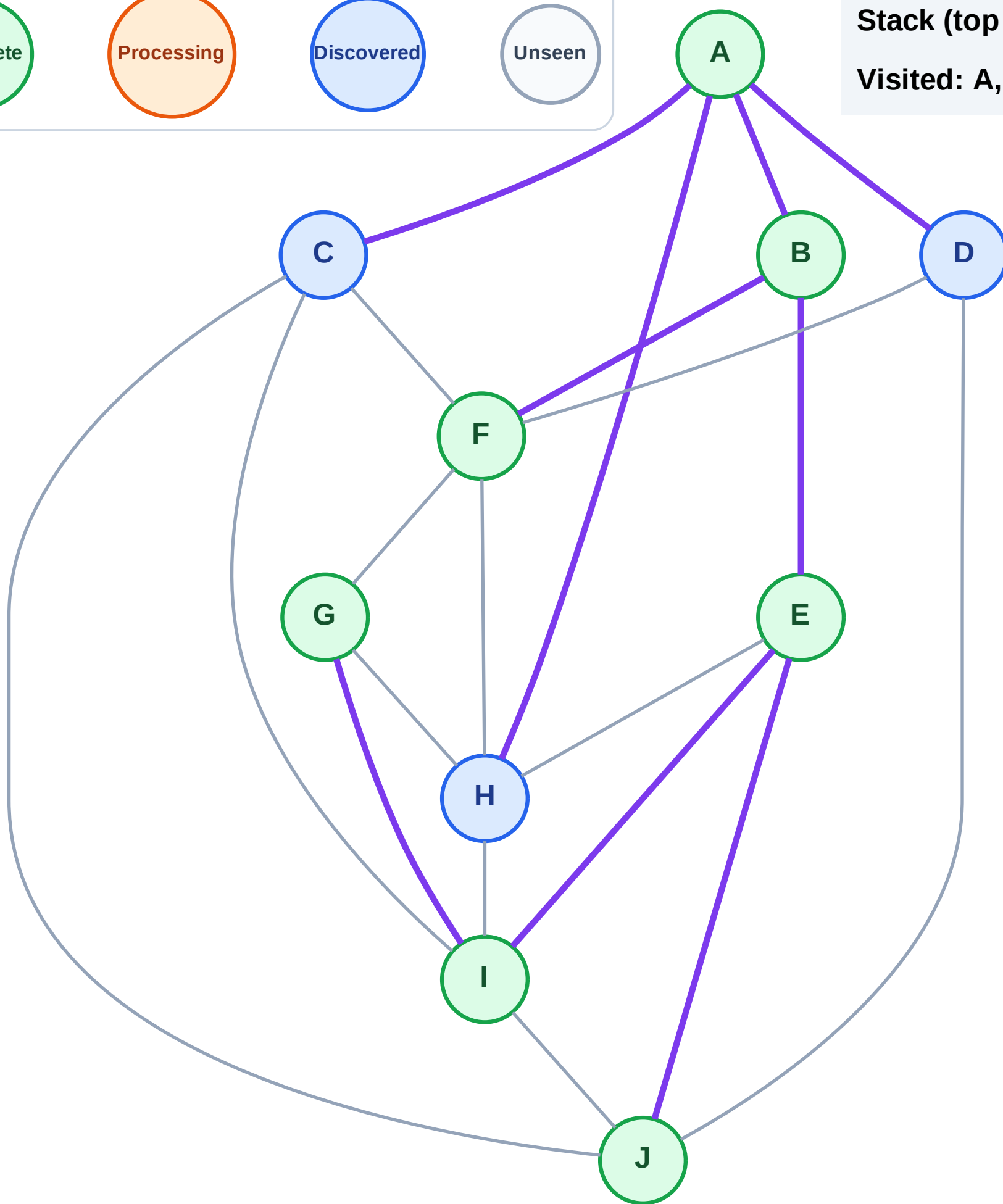
Step 15: No new neighbors from F

Legend



Stack (top at right): H | D | C

Visited: A, B, E, F, G, I, J



DFS Traversal

Step 16: Process node C

Legend

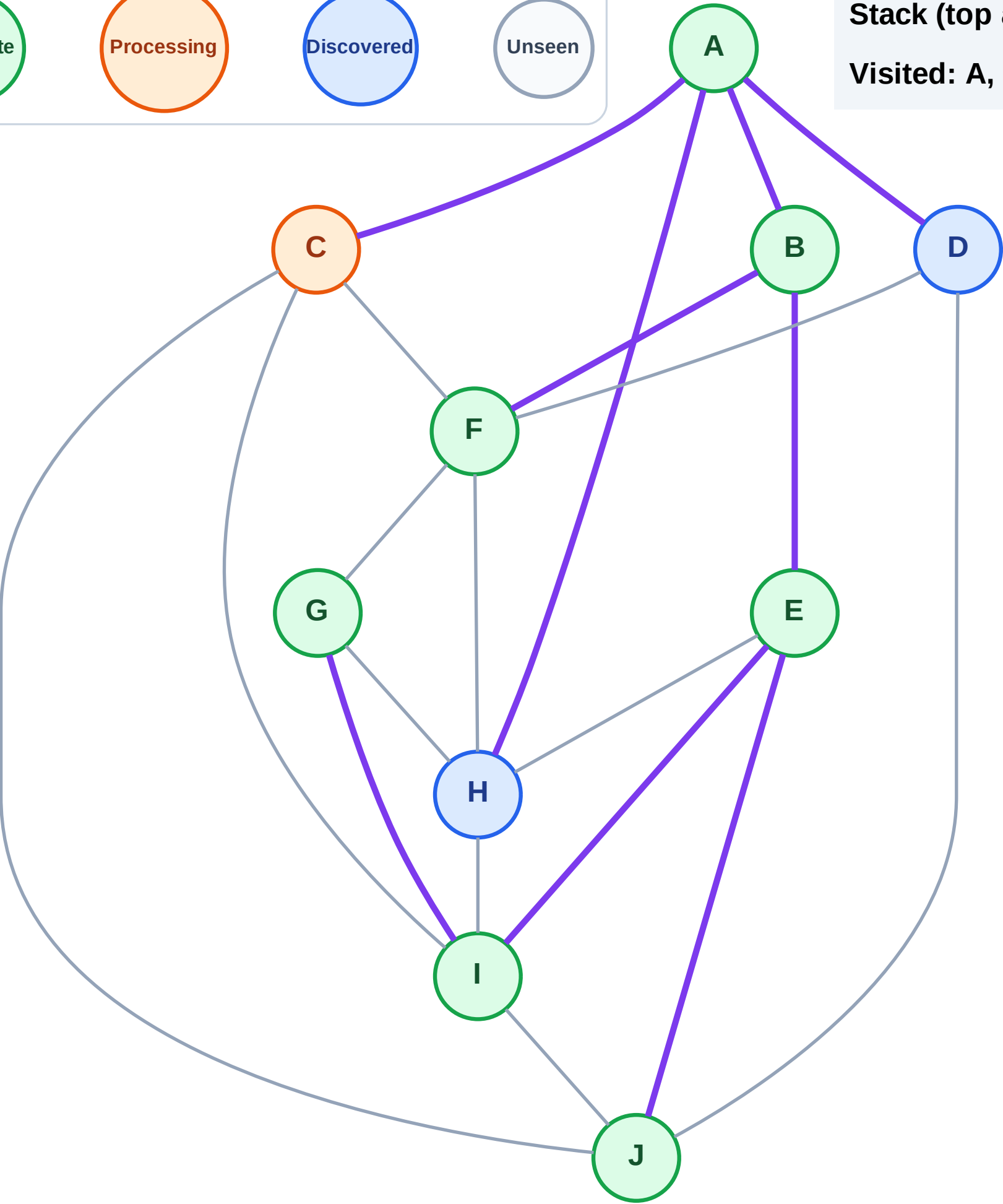
Complete

Processing

Discovered

Unseen

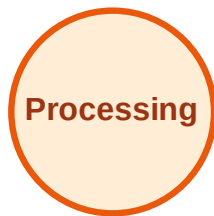
Stack (top at right): H | D
Visited: A, B, E, F, G, I, J



DFS Traversal

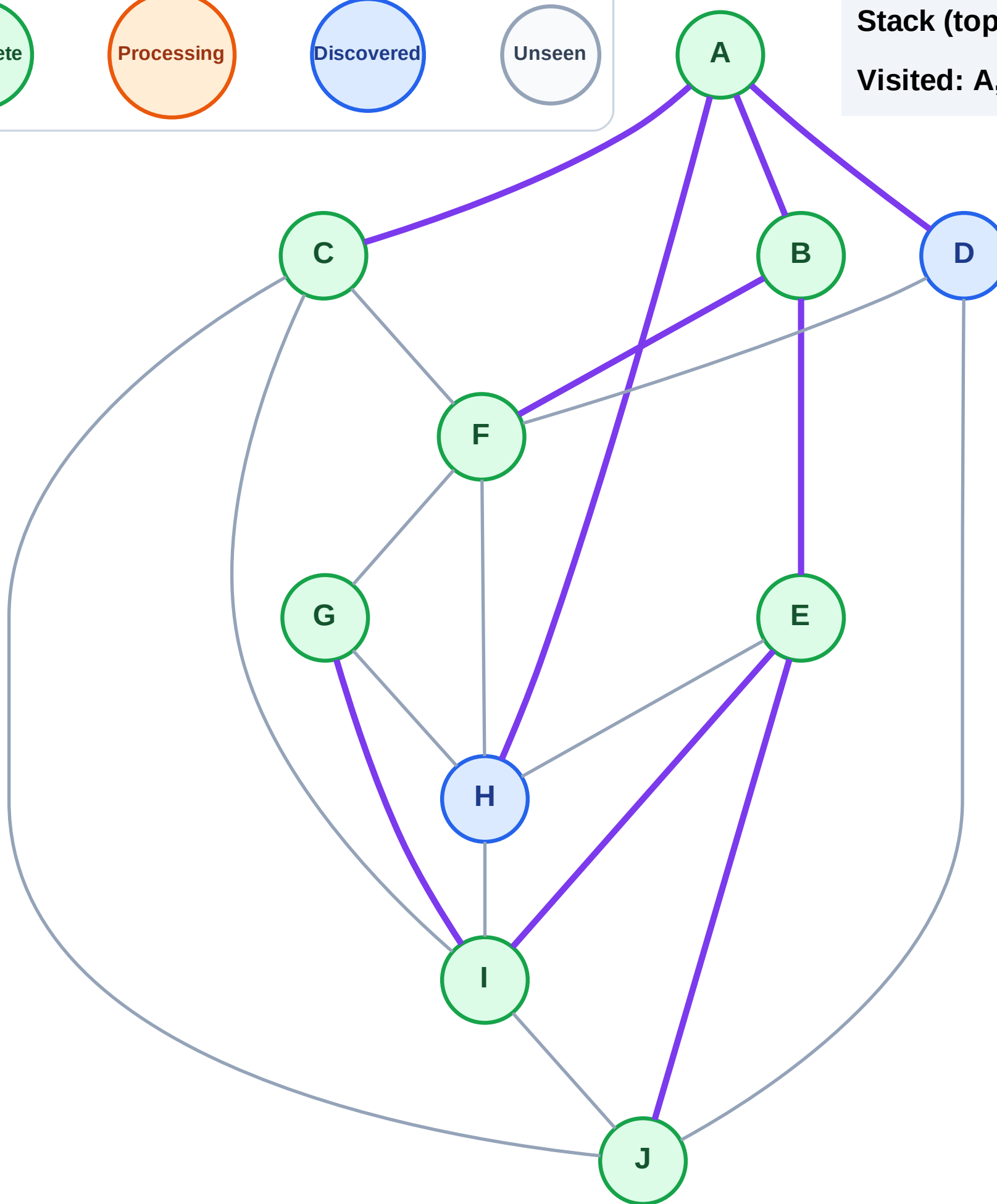
Step 17: No new neighbors from C

Legend



Stack (top at right): H | D

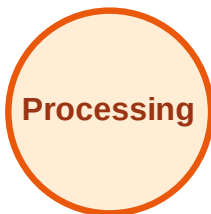
Visited: A, B, C, E, F, G, I, J



DFS Traversal

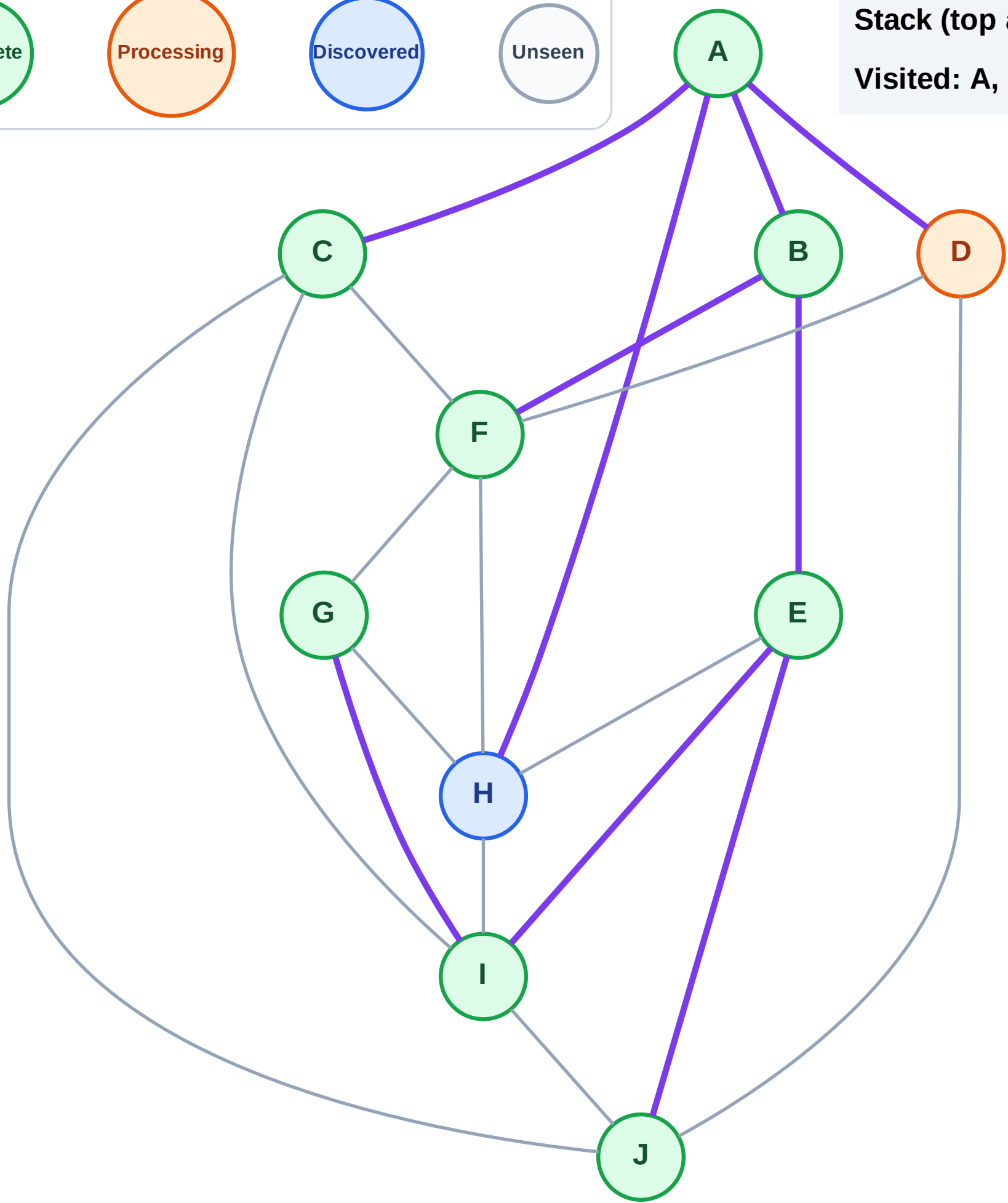
Step 18: Process node D

Legend



Stack (top at right): H

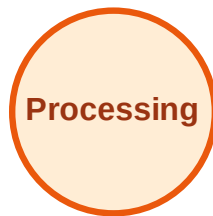
Visited: A, B, C, E, F, G, I, J



DFS Traversal

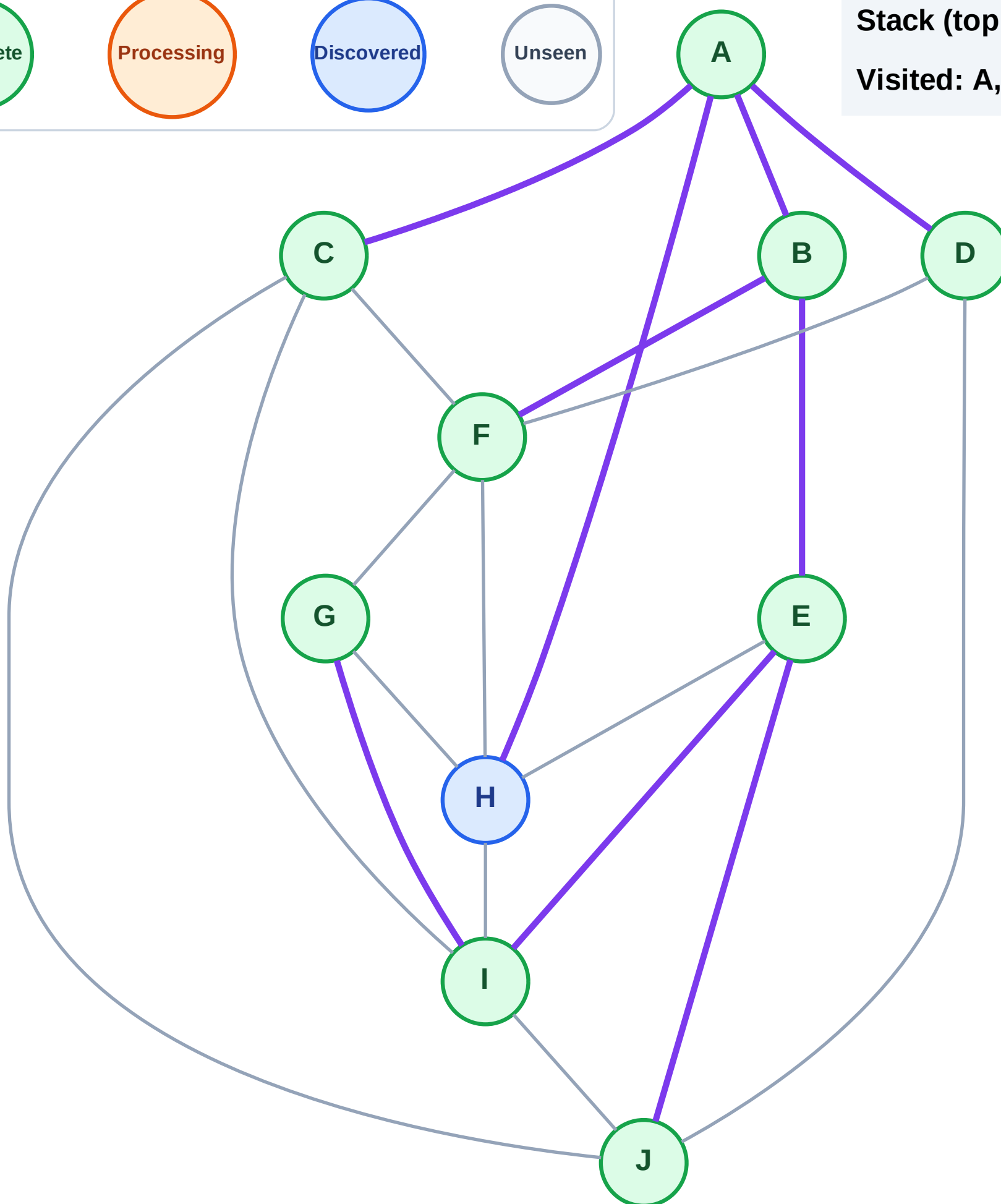
Step 19: No new neighbors from D

Legend



Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 20: Process node H

Legend

Complete

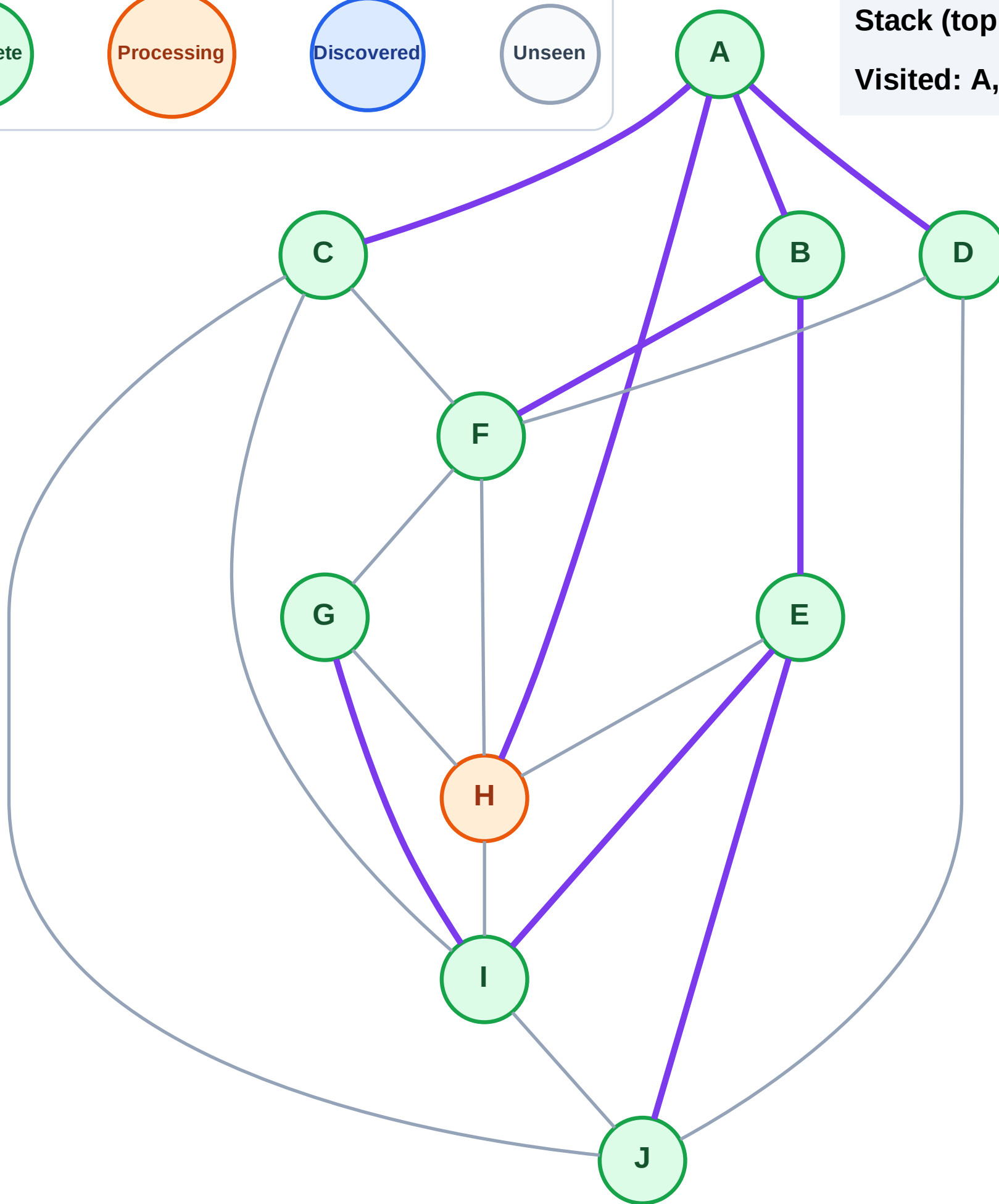
Processing

Discovered

Unseen

Stack (top at right): (empty)

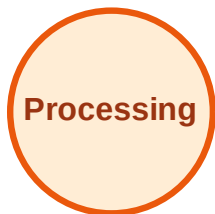
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

